

Modeling oxygen-void interactions in uranium nitride

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Abstract

Oxygen impurities in uranium nitride (UN) are reported to influence its swelling behavior under irradiation, yet the underlying mechanism remains unknown. In this work, we develop a first-principles model that quantifies the interaction of oxygen with voids and fission gas bubbles in UN, leading to a reduction in surface energy that can promote swelling. The analysis reveals that segregation of substitutional oxygen at surface nitrogen sites is the primary driver of surface energy reduction, $|\Delta\sigma|$, while oxygen in surface hollow sites plays a minor and sometimes counteracting role. $|\Delta\sigma|$ is most pronounced for small cavities ($R_v = 1\text{--}10$ nm) at intermediate temperatures that coincide with the onset of breakaway swelling in UN. Larger voids require higher temperatures for oxygen adsorption to significantly lower their surface energy. The temperature dependence of $|\Delta\sigma|$ exhibits three regimes: negligible reduction at low temperatures due to sluggish oxygen diffusion, a maximum at intermediate temperatures where oxygen incorporation is optimal, and a decline at high temperatures due to enhanced bulk solubility. A parametric analysis reveals that $|\Delta\sigma|$ depends strongly on both oxygen concentration and cavity size, but only weakly on porosity. Our results suggest that oxygen-induced surface energy reduction is essential for reconciling the mechanistic swelling model of UN with experimental observations.

Keywords: Uranium nitride, Oxygen segregation, Surface energy reduction, Void nucleation, Fission gas swelling

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1. Introduction

Uranium nitride (UN) is considered a promising fuel for fast nuclear reactors, space reactors, and potentially commercial light water reactors, due to its high uranium density and high thermal conductivity relative to UO_2 [1, 2]. However, early use of UN was limited by issues such as rapid oxidation in air, poor tolerance to water and steam, difficulty in sintering, and the need for nitrogen enrichment with N-15 due to the high neutron absorption of N-14 [1, 2]. Furthermore, many aspects of its behavior at high temperatures and under irradiation remain poorly understood, with scarce experimental data and incomplete mechanistic pictures [1, 2]. Several studies have sought to address this knowledge gap by, for instance, offering an atomistic-level understanding of the deformation mechanisms in UN [3, 4], which play a critical role in pellet-clad mechanical interactions. Additionally, a mechanistic multiscale model has been developed to describe its swelling behavior [5].

UN experiences pronounced swelling driven by the growth of fission gas bubbles to sizes larger than those observed in UO_2 [5–7], making dimensional stability a key concern for this fuel. The mechanistic model by Rizk *et al.* [5] gives a multiscale description of fission-gas swelling and release in UN, coupling atomistic defect data to the BISON finite-element framework via the Simple Integrated Fission Gas Release and Swelling (SIFGRS) model. Their calculations, however, employed a constant void/bubble surface energy derived for oxygen-free UN.

A major challenge in utilizing UN as a nuclear fuel is the presence of oxygen impurities, which can markedly affect its swelling characteristics. Rogozkin *et al.* [8] demonstrated that fuel swelling becomes substantially more pronounced when the oxygen concentration in UN surpasses the range of 1000 to 1500 parts per million (ppm). Consequently, Schuler *et al.* [9] recommended limiting the maximum permissible oxygen content to 1000–1500 ppm to maintain dependable performance of fuel elements. Nevertheless, achieving UN powders with oxygen levels below 1500 ppm has thus far only been accomplished under controlled laboratory conditions [9]. This highlights the importance of investigating how oxygen interacts with UN’s microstructural features, like voids and bubbles.

Additionally, mechanistic models that aim to predict fission gas swelling in UN require knowledge of surface energies. Surface energy is a core input to the SIFGRS model utilized by Rizk *et al.* [5] as it sets both the equilibrium pressure inside a bubble as a function of its radius, and the contact angle of intergranular bubbles. Therefore, understanding whether oxygen significantly alters the surface energy of voids or gas bubbles is essential for accurately modeling swelling behavior.

The effect of light impurities, such as oxygen, on vacancy cluster morphology has been extensively studied in metals like copper, nickel, and stainless steel [10–13]. These studies have utilized elastic continuum models [10, 11], thermodynamic models [12], and reaction rate theory [13] to understand the interaction between impurities and void embryos among other vacancy cluster types. Usually, oxygen impurities impact swelling in metals and alloys by segregating around void surfaces [14], lowering their surface energy, thereby reducing the critical void radius in the void nucleation stage [10, 13]. Experimental evidence from metal studies supports the notion that oxygen reduces the surface energy of small voids, typically around 50 vacancies [12]. This reduction in surface energy is a key factor in stabilizing voids and fission-gas bubbles and may similarly affect the swelling behavior in UN. More

strikingly, the theoretical studies by Zinkle *et al.* [10, 11] predict that void formation in high-purity metals is energetically unfavorable, and impurities like oxygen are essential for voids to form. That is, void formation is greatly reduced or even suppressed in low-oxygen metals and alloys. It has been further suggested that oxygen might stabilize void embryos by preventing their collapse into dislocation loops, even if it does not directly reduce surface energy [11, 12].

Several first-principles studies have explored the properties of oxygen impurities in bulk UN and on its surfaces, calculating the oxygen incorporation, migration, and adsorption energies [9, 15–24]. These studies have demonstrated that oxygen atoms prefer to incorporate into pre-existing nitrogen vacancies, both in bulk UN and on the surface, rather than occupying tetrahedral interstitial sites [15, 21]. Oxygen impurities also reduce the migration energy of nearby nitrogen vacancies [16], indicating that oxygen can influence behaviors that depend on defect dynamics, such as creep. Additionally, it has been observed that O₂ molecules parallel to the (001) surface can spontaneously dissociate when centered over a hollow site or a nitrogen atom [17].

A particularly relevant study on the effect of oxygen on swelling in UN is that by Schuler *et al.* [9], who investigated the thermodynamic and transport properties of oxygen in UN using a self-consistent mean-field (SCMF) model. The input energies for their model were derived from density functional theory with the addition of a Hubbard U term (DFT+ U). Schuler *et al.* found that oxygen is thermodynamically more favorable in the bulk of UN rather than on the (001) surface, leading them to conclude that oxygen segregation on surfaces would be driven by kinetic factors rather than by thermodynamics. However, Kocovski *et al.* [25] later demonstrated that UN modeled using DFT+ U exhibits imaginary phonon frequencies, indicating a dynamically unstable crystal structure. Consequently, the defect energetics used in the SCMF model by Schuler *et al.* might need reconsideration.

Oxygen occurs in fresh UN fuel in solution or as a separate UO₂ phase, dependent on O concentration and temperature [26]. UO₂ precipitates have been observed experimentally in regions of UN samples where the local oxygen concentration apparently exceeded the solubility limit [27]. Additionally, DFT studies indicate that oxygen interaction with UN surfaces can lead to the formation of oxynitrides (UO _{x} N _{y}) or pseudo-UO₂ structures [21, 22].

In this work, we use DFT data to parametrize a thermodynamic model based on defect reactions to study how oxygen impurities interact with voids and fission-gas bubbles in UN. Our model focuses on oxygen adsorption onto void surfaces, where it may reduce surface energy and potentially influence swelling. Throughout this work, we assume that inert gas atoms have no impact on surface energy; therefore, the surface energies of voids and fission-gas bubbles are considered identical. To the best of our knowledge, this is the first study that attempts to quantify the extent to which oxygen interaction with void surfaces affects the void nucleation behavior in UN. We assume that oxygen is the only impurity present, and do not account for the formation of UO₂. These simplifications allow us to concentrate on the primary thermodynamic factors that govern the interaction of oxygen with voids, providing an initial step toward understanding the role of oxygen in UN swelling.

2. Methodology

2.1. DFT calculations

DFT calculations performed in this work utilize the Vienna Ab-initio Simulation Package (VASP) [28–30] using the Perdew-Burke-Ernzerhof (PBE) [31] generalized gradient approximation (GGA) to the exchange-correlation (XC) functional. The pseudopotential models of uranium, nitrogen, and oxygen are based on the projector augmented wave (PAW) method [32] where the valence electron configuration of U is $6s^2 6p^6 6d^2 5f^2 7s^2$ (14 electrons), that of N is $2s^2 2p^3$ (5 electrons), and that of O is $2s^2 2p^4$ (6 electrons). Methfessel and Paxton’s smearing method [33] of the first order is used with a width of 0.1 eV to determine the partial occupancies for each wavefunction.

UN adopts the NaCl structure (space group $Fm\bar{3}m$). $3 \times 3 \times 3$ supercells of ferromagnetic (FM) UN, containing 108 formula units (216 atoms), are used to model bulk UN and to study the properties of defects and impurities. The relaxed lattice parameter is $a_0 = 4.861$ Å with a site-projected uranium magnetic moment of $\mu_U = 1.21\mu_B$, reproducing the PBE FM benchmarks of Kocovski *et al.* [25] and deviating by -0.6% from the experimental value of 4.89 Å [34].

To study surface properties, 6-layer and 8-layer symmetric slabs based on the same 3×3 surface unit cell are used, containing 216 and 288 atoms, respectively, where the vacuum gap in each is twice the length of the slab along the z -direction. It was shown by Kocovski *et al.* [25] that the FM ordering provides a suitable model of UN at 0 K. Brillouin zone sampling is performed using Monkhorst-Pack [35] $3 \times 3 \times 3$ and $3 \times 3 \times 1$ k -point meshes for the $3 \times 3 \times 3$ supercells and the symmetric slabs, respectively. In all energy minimization calculations, we set the cutoff energy of the plane-wave basis as 520 eV, the electronic relaxation convergence criterion as 10^{-4} eV, and the ionic relaxation convergence criterion as 10^{-2} eV/Å. The nudged elastic band (NEB) method [36] with the climbing image algorithm [37] is used to calculate the migration energies of O_i and $\{O_N:v_N\}$. The structures and ionic minimization trajectories are visualized using OVITO [38] and VESTA [39].

Point defect formation energies are calculated from:

$$E_f = E_d - E_p - \sum_i n_i \mu_i, \quad (1)$$

where E_d and E_p are the energies of the supercell with and without the defect, respectively, n_i is the number of atoms of type i removed ($n_i < 0$) or added ($n_i > 0$) to form the defect, and μ_i is the chemical potential of the i th species.

The U and N chemical potentials are calculated from [40]:

$$E_c(U_xN_y) = x\mu_U + y\mu_N, \quad (2)$$

where $E_c(U_xN_y)$ is the cohesive energy (energy per formula unit) of compound U_xN_y . Eq. (2) is solved for UN and α -U to get the U-rich chemical potentials, and for UN and α -U₂N₃ to get the N-rich chemical potentials [41, 42]. The chemical potentials at the intermediate near-stoichiometric conditions are the averages of these two bounds [43].

The binding energy of a defect cluster $\{A_X:B_Y\}$ is calculated from:

$$E_b = E(\{A_X:B_Y\}) + E_p - E(A_X) - E(B_Y), \quad (3)$$

where $E(\cdot)$ is the energy of the supercell containing the relevant defect. Based on this formulation, a negative E_b means binding is favorable.

The incorporation energy of an impurity B in a defect v_X to form B_X is calculated from:

$$E_{\text{inc}} = E(B_X) - E(v_X) - \mu_B, \quad (4)$$

where μ_B is the chemical potential of the impurity in its reference state. The oxygen chemical potential is discussed in [Section 2.2](#). Note that $E(v_i) = E_p$, where v_i denotes a vacant interstitial site.

The surface energy is calculated from [\[44\]](#):

$$\sigma = \frac{1}{A} \left(E^* - \sum_i N_i \mu_i \right), \quad (5)$$

where E^* is the total DFT energy of the supercell containing the surface, A is the total surface area, N_i is the number of atoms of species i in the supercell containing the surface, and μ_i is the corresponding chemical potential. Note that $i \in \{\text{U, N, O}\}$.

The adsorption energy is calculated from:

$$E_{\text{ad}} = \frac{1}{N_{\text{ad}}} (E_{\text{surf+ad}} - E_{\text{surf}}) - \mu_{\text{ad}}, \quad (6)$$

where $E_{\text{surf+ad}}$ and E_{surf} are the energies of the symmetric slabs with and without the adsorbed atom(s), N_{ad} is the number of adsorbed atoms, and μ_{ad} is their chemical potential.

Finally, it is of interest to estimate the change in the surface energy upon oxygen adsorption or surface defect formation. This is accomplished by using the following expression, which is derived in [Appendix B](#) (using geometric identities derived in [Appendix A](#)):

$$\Delta\sigma = \frac{8}{3} \frac{1-p}{p} \frac{R_v}{a^3} c_{\text{O}} \left[\alpha_1 \frac{[\text{O}_{\text{N}}^{(\text{s})}]}{[\text{O}_{\text{N}}^{(\text{b})}]} E_{\text{ad}}(\text{O}_{\text{N}}^{(\text{s})}) + \alpha_2 \frac{[\text{O}_i^{(\text{s})}]}{[\text{O}_{\text{N}}^{(\text{b})}]} E_{\text{ad}}(\text{O}_i^{(\text{s})}) \right] + \frac{2}{a^2} [v_{\text{N}}^{(\text{s})}] E_f(v_{\text{N}}^{(\text{s})}), \quad (7)$$

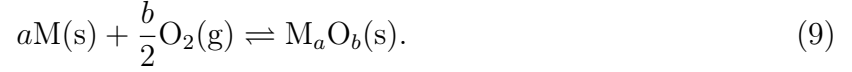
where p is the porosity, R_v is the average void radius, a is the lattice constant, c_{O} is the atomic oxygen concentration, and α_1 and α_2 are the kinetic corrections for the defects $\text{O}_{\text{N}}^{(\text{s})}$ and $\text{O}_i^{(\text{s})}$, respectively. Note that [Eq. \(7\)](#) predicts that the surface energy reduction scales linearly with oxygen concentration. For a defect i , α_i is defined as [\[12\]](#):

$$\alpha_i = \begin{cases} \sqrt{D_i t} / \lambda_v & \text{if } \sqrt{D_i t} / \lambda_v < 1, \\ 1 & \text{if } \sqrt{D_i t} / \lambda_v \geq 1, \end{cases} \quad (8)$$

where D_i is the diffusivity of the relevant defect, t is the void nucleation time, and λ_v is the mean free path between voids, defined as $\lambda_v = n_v^{-1/3}$, where $n_v = p / (\frac{4}{3}\pi R_v^3)$ is the void number density. We assume a void nucleation time $t = 42$ hours. This estimate is justified by taking a typical dose rate, $\dot{G}$, in a fast reactor core as on the order of 10^{-6} dpa/s [\[45, 46\]](#). In stainless steels, stable voids are fully formed by a dose, G , in the range of 0.1–0.2 dpa [\[47\]](#). Using the relation $t = G / \dot{G}$, this corresponds to a void nucleation time between 28 and 56 hours, with an average of 42 hours. Due to the lack of experimental data on early-stage void nucleation in UN fuels, this estimated time serves as a reasonable placeholder. A discussion of the model sensitivity to the kinetic parameter is given in [Section 4](#).

2.2. Oxygen chemical potential

DFT calculations with GGA exchange-correlation functionals are known for being inaccurate for gases like O_2 [21]. This is usually remedied by using hybrid functionals or applying semi-empirical corrections [48]. Alternatively, Finnis *et al.* [44] suggested using common oxides as the oxygen reference state. For an oxide M_aO_b , the formation reaction is:



The Gibbs free energy balance for this reaction is:

$$\Delta_f G^0(M_aO_b) = g^0(M_aO_b) - a\mu_M^0 - b\mu_O^0, \quad (10)$$

where $\Delta_f G^0(M_aO_b)$ is the standard Gibbs free energy of formation for the oxide, which is available in thermochemical tables [49], $g^0(M_aO_b)$ is approximated by the DFT energy per formula unit of the oxide, μ_M^0 is approximated by the DFT energy per atom of the metal, and μ_O^0 is the oxygen chemical potential. The superscript 0 on any quantity denotes its value at the standard pressure, $P^0 = 1$ bar, and standard temperature, $T^0 = 298.15$ K. Approximating T^0 with 0 K in the context of DFT calculations introduces little error [44]. Rearranging Eq. (10), the oxygen chemical potential is:

$$\mu_O^0 = \frac{1}{b} [g^0(M_aO_b) - a\mu_M^0 - \Delta_f G^0(M_aO_b)]. \quad (11)$$

Finnis *et al.* state that the chemical potential of oxygen should be independent of the specific oxide used. Using MgO and Al_2O_3 as the initial oxide, we found that the chemical potential for both is $\mu_O = -4.18$ eV. This value is used as the reference chemical potential to report all the oxygen incorporation and adsorption energies in Section 3. However, we are concerned with studying how oxygen interacts with voids in UN, where oxygen atoms are introduced as impurities during manufacturing. For this setting, the oxygen reference state is not the oxide, but rather the most dominant oxygen impurity type, which is $O_N^{(b)}$ as will be shown in Section 3. Note that we use the Kröger-Vink notation [50] to describe point defects, but with charges omitted because UN exhibits metallic properties and defects have no associated charge [51]. Superscripts (b) and (s) are added to the Kröger-Vink notation to differentiate bulk and surface defects, respectively.

The reference state of oxygen impurities in UN is based on the following defect reaction:



which has the following mass-action law:

$$K_1 = \frac{[O_N^{(b)}]}{[v_N^{(b)}]P_{O_2}^{1/2}} = \exp\left[-\frac{E(O_N) - E(v_N)}{kT}\right], \quad (13)$$

where $E(O_N)$ and $E(v_N)$ are the DFT energies of the supercells containing O_N and v_N , respectively. This difference is termed the “raw incorporation energy” [40], i.e., the incorporation

energy without considering the impurity's reference state. The dependence of the oxygen chemical potential on oxygen partial pressure, P_{O_2} , and temperature, T , is given by [44]:

$$\mu_{\text{O}}(P_{\text{O}_2}, T) = g(T) + \frac{1}{2}kT \ln(P_{\text{O}_2}/P^0), \quad (14)$$

where the temperature dependence, $g(T)$, is extracted from thermochemical tables [49]. Focusing on the dependence on the oxygen partial pressure:

$$\mu_{\text{O}}(P_{\text{O}_2}) = \frac{1}{2}kT \ln(P_{\text{O}_2}/P^0), \quad (15)$$

and rearranging Eqs. (13) and (15), it can be easily shown that the oxygen chemical potential is:

$$\mu_{\text{O}} = E(\text{O}_{\text{N}}) - E(v_{\text{N}}) + kT \ln\left([\text{O}_{\text{N}}^{(\text{b})}]/[v_{\text{N}}^{(\text{b})}]\right). \quad (16)$$

2.3. Oxygen diffusivity

The diffusivity of oxygen impurities is also explored in this work. The diffusivity of a defect d is calculated from:

$$D_d = \frac{1}{6}z\lambda^2\nu \exp\left(-\frac{E_m}{kT}\right), \quad (17)$$

where $d \in \{\text{O}_i, \{\text{O}_{\text{N}}:v_{\text{N}}\}\}$, z is the number of equivalent sites the atom can jump to, and λ is the jump distance. Both z and λ are determined from the crystal structure. For O_i , $z = 6$ and $\lambda = a/2$, with a being the lattice constant. For $\{\text{O}_{\text{N}}:v_{\text{N}}\}$, $z = 12$ and $\lambda = a/\sqrt{2}$. E_m is the effective migration energy. For O_i , $E_m = 2.41$ eV. For $\{\text{O}_{\text{N}}:v_{\text{N}}\}$, net bulk diffusion requires two steps [52]: (a) O_{N} and v_{N} exchange, and (b) v_{N} rotation around O_{N} . Note the O_{N} is immobile by itself and can only move via this two-step process with the assistance of a nearby v_{N} . The saddle point energy of the exchange step is 2.87 eV, whereas that of the rotation step is 3.06 eV. Based on the highest barrier approximation [53], the effective migration energy is 3.06 eV.

In Eq. (17), ν is the attempt frequency, which can be estimated from phonon calculations. Because these calculations are computationally expensive, we use a semi-classical approach to estimate ν as [54]:

$$\nu = \left(\frac{E_m}{2m\lambda^2}\right)^{1/2}, \quad (18)$$

where m is the mass of the diffusing atom. For $\{\text{O}_{\text{N}}:v_{\text{N}}\}$, the limiting step is the rotation of v_{N} around O_{N} and the mass of the nitrogen atom is used. For O_i , the mass of the oxygen atom is used. For the diffusion of $\{\text{O}_{\text{N}}:v_{\text{N}}\}$, $\nu = 9.45 \times 10^{12}$ Hz, whereas for O_i , $\nu = 1.11 \times 10^{13}$ Hz.

The diffusivity of an impurity X by a mechanism that depends on defect d is calculated from [51]:

$$D_{X,d} = f \frac{c_d}{c_{\text{X}}} D_d, \quad (19)$$

where f is the correlation factor. For $\{\text{O}_{\text{N}}:v_{\text{N}}\}$, $f = 0.7815$ [55], which is the value for vacancy-mediated migration in the FCC nitrogen sublattice. For O_i , $f = 1$ [55]. c_d is the atomic concentration of the defect d and c_{X} is the total atomic concentration of impurity X. As an approximation, we assume that $c_{\text{O}} = [\text{O}_{\text{N}}]$ which will be justified in Section 3.3 of the results.

2.4. Defect reactions

In this section, we outline the defect reactions used to estimate the relative concentrations of various oxygen defect types. In general, there are three categories of defect reactions: (a) reactions that relate defects in the bulk, (b) reactions that relate defects on the void surfaces, and (c) reactions that thermodynamically dictate defect transitions from the bulk to the void surfaces, and vice versa. A limitation of the latter category of reactions is that it neglects the kinetic aspects of the defect transitions. That is, the kinetics of reactions in (c) category are assumed to be faster than those in the (a) and (b) categories.

The first bulk defect reaction is:



and its mass-action law is:

$$K_2 = \frac{[\text{O}_i^{(\text{b})}][v_\text{N}^{(\text{b})}]}{[\text{O}_\text{N}^{(\text{b})}]} = \exp \left[-\frac{E_\text{inc}(\text{O}_i^{(\text{b})}) - E_\text{inc}(\text{O}_\text{N}^{(\text{b})})}{kT} \right]. \quad (21)$$

The second bulk defect reaction is:



and its mass-action law is:

$$K_3 = \frac{[\{\text{O}_\text{N}:v_\text{N}\}^{(\text{b})}]}{[v_\text{N}^{(\text{b})}][\text{O}_\text{N}^{(\text{b})}]} = \exp \left[-\frac{E_\text{b}(\{\text{O}_\text{N}:v_\text{N}\}^{(\text{b})})}{kT} \right]. \quad (23)$$

The defect reaction at void surfaces is:



and its mass-action law is:

$$K_4 = \frac{[v_\text{N}^{(\text{s})}][\text{O}_i^{(\text{s})}]}{[v_i^{(\text{s})}][\text{O}_\text{N}^{(\text{s})}]} = \exp \left[-\frac{E_\text{ad}(\text{O}_i^{(\text{s})}) - E_\text{ad}(\text{O}_\text{N}^{(\text{s})})}{kT} \right]. \quad (25)$$

The first reaction that relates bulk UN and the void surface is:



and its mass-action law is:

$$K_5 = \frac{[v_\text{N}^{(\text{s})}][\text{N}_\text{N}^{(\text{b})}]}{[v_\text{N}^{(\text{b})}][\text{N}_\text{N}^{(\text{s})}]} = \exp \left[-\frac{E_\text{f}(v_\text{N}^{(\text{s})}) - E_\text{f}(v_\text{N}^{(\text{b})})}{kT} \right]. \quad (27)$$

$[\text{N}_\text{N}^{(\text{s})}]$, the atomic concentration of the nitrogen sites on the surface, can be estimated from knowledge of the porosity, p , and the average void radius, R_v by the following expression which is derived in [Appendix A](#):

$$[\text{N}_\text{N}^{(\text{s})}] = \frac{3}{2} \frac{p}{1-p} \frac{a}{R_v}, \quad (28)$$

where a is the lattice constant. Note that $[N_N^{(b)}] = 1 - [N_N^{(s)}]$.

The second reaction is:



and its mass-action law is:

$$K_6 = \frac{[v_i^{(s)}][O_N^{(b)}]}{[v_N^{(b)}][O_i^{(s)}]} = \exp \left[-\frac{E_{\text{inc}}(O_N^{(b)}) - E_{\text{ad}}(O_i^{(s)})}{kT} \right]. \quad (30)$$

As derived in [Appendix A](#), $[v_i^{(s)}]$ is given by:

$$[v_i^{(s)}] = \frac{3p}{1-p} \frac{a}{R_v}. \quad (31)$$

To find the relative concentration $[O_N^{(s)}]/[O_N^{(b)}]$, it is not necessary to resort to a defect reaction. Instead, the ratio is calculated from:

$$\frac{[O_N^{(s)}]}{[O_N^{(b)}]} = \frac{[O_i^{(s)}]}{[O_N^{(b)}]} \div \frac{[O_i^{(s)}]}{[O_N^{(s)}]} = \frac{[v_N^{(s)}]}{[v_N^{(b)}]} \exp \left[-\frac{E_{\text{ad}}(O_N^{(s)}) - E_{\text{inc}}(O_N^{(b)})}{kT} \right]. \quad (32)$$

3. Results

3.1. Defect energetics

The formation energies of U and N vacancies in UN are shown in [Table 1](#). The bulk formation energies compare very well with those calculated by Yang and Kaltsoyannis [42] and Kocovski *et al.* [25], which acts as a validation of our DFT model. It is noticed that there is a consistent difference of about 0.76 eV between N vacancy formation on the surface and in the bulk, independent of stoichiometry. This difference is termed the segregation energy of the N vacancy. It is easier to form an N vacancy on the surface than in the bulk, and the number of $v_N^{(s)}$ sites is limited by the available surface area.

Table 1: Formation energies (eV) of U and N vacancies in UN. Reference energies are DFT values calculated by Yang and Kaltsoyannis [42] and Kocovski *et al.* [25].

	U-rich		Intermediate		N-rich	
	Calc.	Ref.	Calc.	Ref.	Calc.	Ref.
$v_U^{(b)}$	3.28	3.17–3.43 [42], 3.27–3.86 [25]	2.79	2.75–3.01 [42]	2.30	2.34–2.60 [42], 2.09–2.58 [25]
$v_N^{(b)}$	1.64	1.76–1.90 [42], 0.62–1.86 [25]	2.13	2.18–2.31 [42]	2.62	2.59–2.72 [42], 1.42–2.82 [25]
$v_N^{(s)}$	0.88	–	1.37	–	1.86	–

The incorporation energies of oxygen in UN are shown in [Table 2](#), all given relative to $\mu_O = -4.18$ eV. The most favorable oxygen defect in the bulk is O_N followed by O_i . We also attempted to calculate the incorporation energy of the $\langle 111 \rangle$ and $\langle 110 \rangle$ O-N dumbbells. The $\langle 111 \rangle$ O-N dumbbell has a larger incorporation energy (−1.11 eV) than the tetrahedral site, whereas the $\langle 110 \rangle$ O-N dumbbell is unstable and relaxes to the $\langle 111 \rangle$ configuration. The largest incorporation energy is that of O_U , which indicates that O impurities exist almost exclusively in the N-sublattice.

Table 2: Incorporation energies of oxygen in UN for $\mu_{\text{O}} = -4.18$ eV. This value of μ_{O} is calculated according to the Finnis *et al.* [44] oxide method.

Defect	Incorporation energy [eV]
O_{N}	-7.23
O_{i}	-2.94
$\langle 111 \rangle$ O-N dumbbell	-1.11
$\langle 110 \rangle$ O-N dumbbell	Unstable
O_{U}	-0.40

The binding energies of the explored defect clusters are shown in Table 3, where negative values indicate favorable binding. The most relevant defect cluster to our study, i.e., $\{\text{O}_{\text{N}} : v_{\text{N}}\}$, has a binding energy of 0.03 eV, which is practically zero. That is, binding is neither favorable nor unfavorable, and the N vacancy can roam around independently of O_{N} . On the other hand, we found a strong tendency for binding between O_{N} and v_{U} , Kr_{U} , and Xe_{U} , all with an average binding energy of about -0.4 eV. We propose that this binding may promote the initial aggregation of defects into void or bubble embryos. This finding is consistent with experimental observations by Turos *et al.* [56], who reported that Kr and Xe atoms form impurity-defect complexes in UN, serving as precursors for gas bubble formation. A comparable phenomenon is observed with helium in copper [11], where helium does not reduce the void surface energy but stabilizes void embryos by binding with vacancies and vacancy clusters. A similar role may be played by oxygen in UN, potentially contributing to the stabilization of voids and bubbles. Due to the low concentrations of Kr_{U} and Xe_{U} , we anticipate that this effect is less significant than the direct reduction of surface energy through oxygen adsorption.

Table 3: Binding energies of defect clusters in UN. Negative values mean that binding is favorable and vice versa.

Defect cluster	Binding energy [eV]
$\{\text{O}_{\text{N}} : v_{\text{N}}\}$	0.03
$\{\text{O}_{\text{N}} : v_{\text{U}}\}$	-0.40
$\{\text{O}_{\text{N}} : \text{Kr}_{\text{U}}\}$	-0.41
$\{\text{O}_{\text{N}} : \text{Xe}_{\text{U}}\}$	-0.39

The migration energies of defects relevant to this study are shown in Table 4. As can be expected, the migration energy of O_{i} is smaller than that of $\{\text{O}_{\text{N}} : v_{\text{N}}\}$, which is 3.06 eV based on the highest barrier approximation. It is predicted that the migration energy of v_{N} is slightly reduced from 3.10 to 3.06 eV when an O_{N} is in a first nearest neighbor position. A similar observation has been made by Kotomin *et al.* [16].

Table 4: Migration energies of some defects and/or migration paths in UN.

Defect and/or migration path	Migration energy [eV]
v_{N}	3.10
O_{i}	2.41
$\text{O}_{\text{N}}-v_{\text{N}}$ exchange	2.87
v_{N} rotation around O_{N}	3.06

3.2. UN surface properties

Upon relaxation, the 8-layer UN (001) surface undergoes a slight adjustment of the interlayer spacing, with the maximum interlayer spacing change remaining below 1%. The energy of the UN (001) surface under various stoichiometric conditions with and without defects is presented in Table 5. Our calculated pristine surface energy of approximately 1.59 J/m² falls within the range of 1.22–1.70 J/m² reported by Bocharov *et al.* [20]. To our knowledge, no experimental measurements of UN surface energy currently exist, making this computational comparison the only benchmark available. It is evident that the surface energies of the 6-layer and 8-layer slabs are very similar, indicating that the 6-layer slab is sufficient to achieve a converged description of the UN (001) surface. Therefore, the 6-layer slab is used in all subsequent adsorption calculations. Several observations can be made regarding the impact of adsorption on surface energy. Introducing O_i^(s) onto the pure 6-layer surface decreases its energy by 0.182 J/m², irrespective of the stoichiometry. Furthermore, the surface energy of the 6-layer surface with O_N^(s) is reduced relative to that of the 6-layer surface with v_N^(s) by 0.256 J/m², also independent of stoichiometry.

Table 5: Surface energies of the UN (001) surface in units of J/m² as calculated by Eq. (5).

	U-rich	Intermediate	N-rich
6-layer pure surface	1.586	1.586	1.586
8-layer pure surface	1.592	1.592	1.592
6-layer surface with O _i ^(s)	1.404	1.404	1.404
6-layer surface with v _N ^(s)	1.620	1.638	1.656
6-layer surface with O _N ^(s)	1.364	1.382	1.400

The oxygen adsorption sites explored in this work are shown in Fig. 1 generated using VESTA. Oxygen adsorption energies in different sites on the 6-layer slab are given in Table 6. The most stable site is a N surface vacancy, followed by the hollow site, i.e., O_i^(s), as was also found in previous studies [15, 21]. The adsorption energy of the oxygen in the hollow site does not change if the oxygen atoms are on one or both sides of the slab, which indicates the absence of any polarity effects. Thus, considering only one side of the slab is sufficient for obtaining accurate adsorption energies. Oxygen adsorption energy in a hollow site slightly decreases from −4.84 eV to −4.85 eV when an oxygen atom is introduced in a nearest neighbor hollow site and increases to −4.72 eV for 4 neighboring O atoms forming a square, which is due to the lateral repulsive interaction between closely packed oxygen adsorbate atoms. This suggests that the saturation coverage of oxygen is probably smaller than 1 ($\theta_s < 1$). However, in this model, oxygen-oxygen interactions are neglected, and a saturation coverage of 1 is assumed for all impurity sites. If the actual saturation coverage was, say, $\theta_s = 0.5$, the effect of this would be to divide $[v_i^{(s)}]$ by 2, which is negligible on a log scale.

Note that the adsorption energy of O atoms in above-U sites is the same as that of O in a hollow site (Table 6). To confirm that the above-U site is stable, we shifted the oxygen atom by 0.2 Å toward the hollow site, and it relaxed back to the above-U site. One way to account for the above-U sites in our model is to multiply $[v_i^{(s)}]$ by 2, as outlined in Appendix A. The bridge site is marked as unstable since oxygen atoms in this configuration relax to the above-U position. The largest adsorption energy is that of the above-N site (equivalent to a site on the U sublattice).

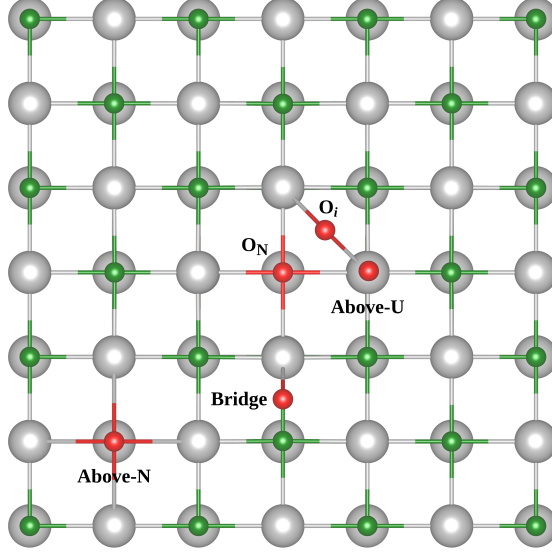


Figure 1: (Color online) A view of the UN surface along the z -axis (i.e., surface normal) showing the different oxygen adsorption sites. U atoms are gray, N atoms are green, and O atoms are red.

Table 6: Adsorption energy of oxygen into sites on the symmetric 6-layer slab model of the UN (001) surface at a coverage of $\theta = 1/9$ ML (one O atom on one side of the 3×3 surface unit cell). These values correspond to $\mu_{\text{O}} = -4.18$ eV, which is calculated according to the Finnis *et al.* [44] oxide method.

Adsorption site	Adsorption energy [eV]
$v_{\text{N}}^{(\text{s})}$	-6.80
Hollow site (one side), i.e., $\text{O}_i^{(\text{s})}$	-4.84
Hollow site (both sides)	-4.84
2 nearest-neighbor hollow sites (dumbbell)	-4.85
4 nearest-neighbor hollow sites (square)	-4.72
Bridge site	Unstable
Above-U site	-4.83
Above-N site	-2.77

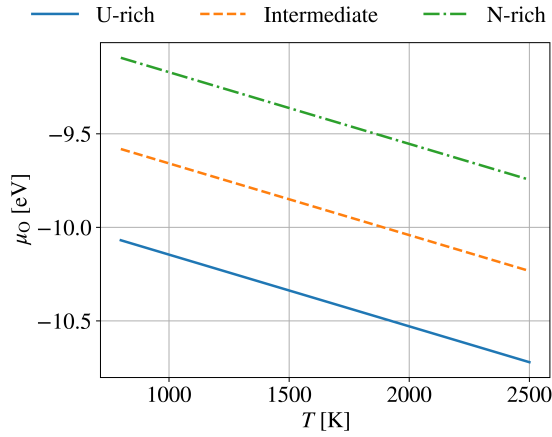


Figure 2: Oxygen chemical potential as calculated based on Eq. (16).

The oxygen adsorption energies in Table 6 are reported with the chemical potential of oxygen taken from the oxide as the reference state (i.e., Eq. (11)). As explained earlier, however, the more appropriate oxygen reference state is oxygen pre-existing in UN as an impurity. In this case, the relevant chemical potential is that shown in Fig. 2. The adsorption energies of oxygen relative to that chemical potential are shown in Figs. 3a and 3b for $O_N^{(s)}$ and $O_i^{(s)}$, respectively. Comparing these adsorption energies to those in Table 6, we can differentiate between two cases: While oxygen adsorption on external surfaces reduces surface energy both as $O_N^{(s)}$ and $O_i^{(s)}$, pre-existing oxygen in UN adsorbing on internal void surfaces reduces the surface energy only when adsorbing as $O_N^{(s)}$. In contrast, $O_i^{(s)}$ increases the surface energy of internal void surfaces.

The energetic preference of $O_N^{(s)}$ over $O_i^{(s)}$ (Table 6) can be understood from the electronic structure analyses of oxygen on UN surfaces reported in [17–19]. In the NaCl structure, a nitrogen sublattice site is coordinated entirely by uranium atoms. Projected density of states (DOS) calculations [17, 19] show that oxygen at such a site on the surface has its $2p$ states partly mixed with U $5f$ states, while the N $2p$ band position remains insensitive to the presence of oxygen. This indicates that the relevant bonding interaction is between oxygen and its nearest-neighbor uranium atoms rather than with nitrogen, consistent with the formation of O-U bonds reported in [19]. Bader charge analysis [19] confirms a substantial charge transfer to the oxygen atom in the $O_N^{(s)}$ site. The hollow interstitial site, by contrast, places oxygen between both U and N surface atoms rather than exclusively among uranium nearest neighbors (see Fig. 1). Since the nitrogen neighbors do not contribute significantly to bonding, the reduced uranium coordination at the hollow site results in weaker O $2p$ -U $5f$ hybridization compared to $O_N^{(s)}$. Charge density difference maps [19] confirm that the bonding perturbation around oxygen in a nitrogen site is spatially localized and directed along the O-U bonds. These findings establish that the O $2p$ -U $5f$ hybridization and charge transfer enabled by the uranium coordination shell at nitrogen sublattice sites is the electronic origin of the stronger adsorption of $O_N^{(s)}$ and, consequently, its dominant role in reducing the surface energy of voids in UN.

3.3. Oxygen behavior

Several parameters must be defined to study the behavior of oxygen using our model. We assume an initial oxygen concentration of $w_O = 1500$ ppm, a porosity of $p = 5\%$, and an average void radius of $R_v = 1$ nm. The chosen $w_O = 1500$ ppm is the upper limit of the tolerated oxygen impurity concentration in UN [8, 9]. The optimal initial (closed) porosity for UN fuels to both suppress oxidation and accommodate gaseous fission products is estimated to be 4% [57]. Note that the porosity in irradiated fuels is typically as high as 15% [58], which exceeds the 5% porosity assumed here. Lastly, the lower limit of reported fission gas bubble sizes in nitride and carbide fuels is close to $R_v = 1$ nm [59], which is also the critical void radius in metals [11].

The atomic concentrations of the N vacancy in the bulk and on the void surfaces are calculated with the aid of formation energies in Table 1 and Eq. (27) and are shown in Figs. D.9a and D.9b, respectively. The concentrations of $O_i^{(b)}$ and $\{O_N : v_N\}^{(b)}$, relative to $O_N^{(b)}$ are shown in Figs. 4a and 4b, respectively. It is clear that $O_N^{(b)}$ is the dominant defect type for oxygen impurities. Thus, it is justified to set $[O_N^{(b)}] \approx c_O$ where c_O is the atomic

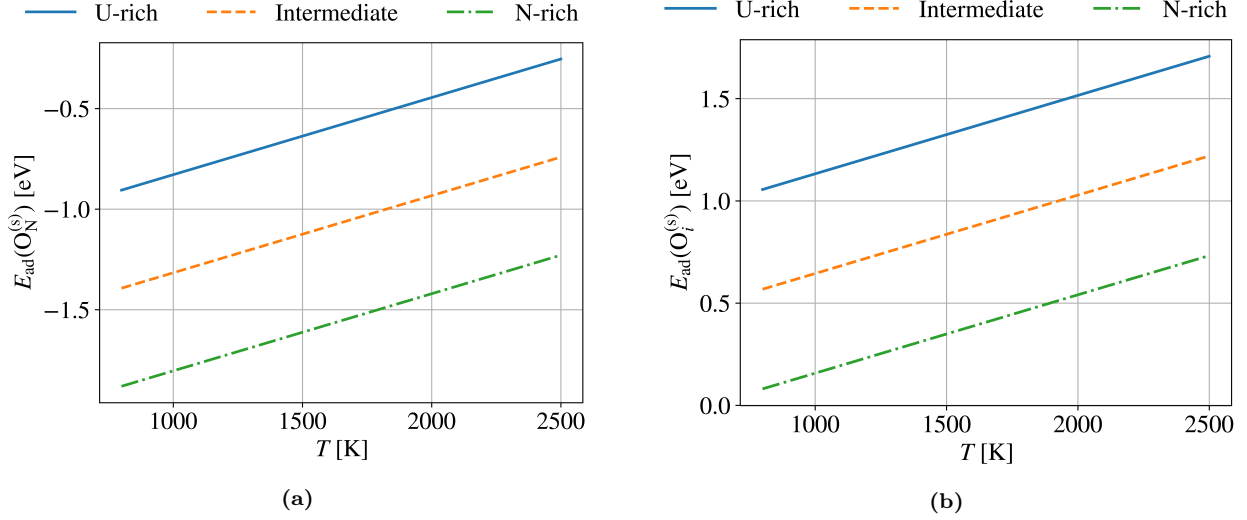


Figure 3: Temperature evolution of oxygen adsorption energy relative to the chemical potential shown in Fig. 2 for (a) $O_N^{(s)}$ and (b) $O_i^{(s)}$.

fraction of oxygen impurities, given by Eq. (B.14). $O_N^{(b)}$ is an immobile defect. Surface energy reduction requires O migration to void surfaces, which can only occur via the mobile defects $O_i^{(b)}$ and $\{O_N:v_N\}^{(b)}$. However, the immobile $O_N^{(b)}$ defects can stabilize void embryos via another mechanism by binding to v_U , Kr_U , and Xe_U , as explained earlier.

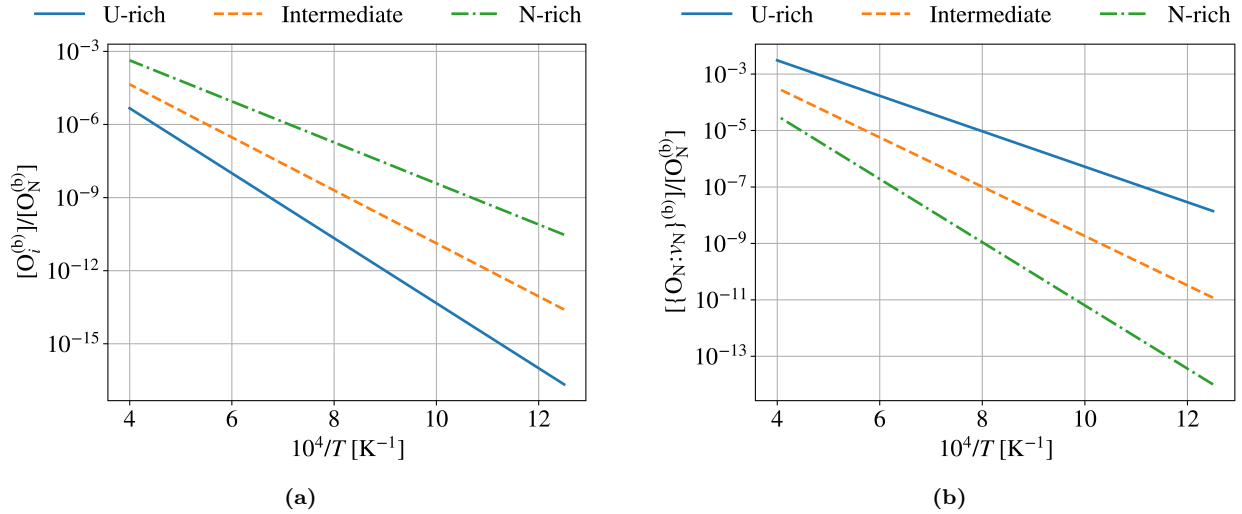


Figure 4: Relative concentrations of (a) $O_i^{(b)}$ and (b) $\{O_N:v_N\}^{(b)}$.

The relative concentration $[O_i^{(s)}]/[O_N^{(b)}]$ is illustrated in Fig. 5a. Under U-rich and intermediate conditions, this relative concentration exhibits Arrhenius behavior and remains relatively low. However, under N-rich conditions, which are more relevant in irradiation environments, the relative concentration is significantly higher and follows an anti-Arrhenius trend. Around 1600 K, for instance, for every ten oxygen atoms in the bulk, approximately one oxygen atom segregates to the surface as $O_i^{(s)}$. This behavior arises because, in N-rich

environments, there are fewer nitrogen vacancies available to form $O_N^{(s)}$, leading to a preference for oxygen atoms to segregate to the surface. Similarly, in Fig. 5b, an anti-Arrhenius trend is also observed for the relative concentration $[O_N^{(s)}]/[O_N^{(b)}]$. This behavior is independent of stoichiometry, as it is solely governed by the segregation energy. As expected, this anti-Arrhenius behavior indicates that the solubility of oxygen in the bulk increases with increasing temperature. Lastly, Fig. 5c depicts the relative concentration $[O_i^{(s)}]/[O_N^{(s)}]$. As expected, $O_N^{(s)}$ is the dominant oxygen impurity site on the void surfaces. Interestingly, despite the large difference in adsorption energies, this relative concentration exceeds 0.1 in N-rich conditions across all considered temperatures.

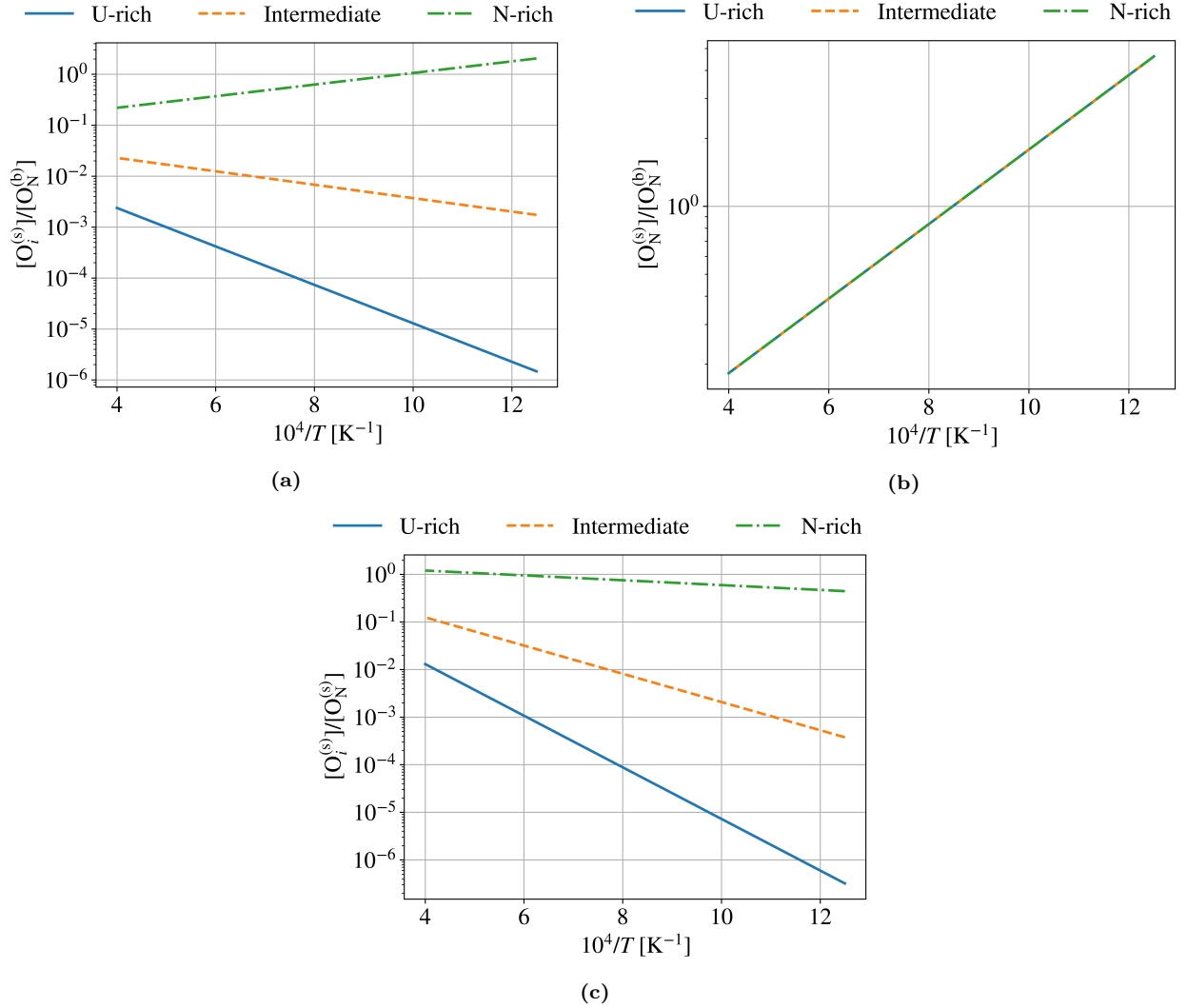


Figure 5: Relative concentrations of (a) $O_i^{(s)}$ and (b) $O_N^{(s)}$, as well as (c) the ratio of their concentrations.

The diffusivity of oxygen impurities in bulk UN through mechanisms dependent on the $\{O_N : v_N\}$ and O_i defects is illustrated in Fig. D.10. Under U-rich conditions, the defect cluster $\{O_N : v_N\}$ dominates oxygen diffusion. This can be explained by the fact that U-rich environments have a higher concentration of nitrogen vacancies, allowing oxygen to

exist as O_N in the bulk. These nitrogen vacancies also facilitate the diffusion process. In intermediate conditions, both diffusion mechanisms are comparable, although O_i exhibits slightly higher diffusivity. In contrast, under N-rich conditions, the availability of nitrogen vacancies decreases, and oxygen predominantly exists as an interstitial O_i . Consequently, the diffusion mechanism via O_i becomes dominant in this regime.

It is important to note that we assume $O_N^{(s)}$ defects reach the surface predominantly through the diffusion of $\{O_N : v_N\}$. This assumption is later utilized when calculating the reduction in surface energy due to surface adsorption. Specifically, the diffusivity used to calculate α_1 in Eq. (7) is that of $\{O_N : v_N\}$.

The variation in surface energy due to oxygen adsorption and vacancy formation on void surfaces is depicted in Fig. 6 for various porosities, oxygen concentrations, and average void radii. At low temperatures, the change in surface energy is negligible. This is because oxygen diffusivity, incorporated in the model via the kinetic correction, α_i , is very slow at low temperatures. Within the void nucleation time, the mobility of oxygen impurities is insufficient for them to reach the void surfaces. For the reference case of $w_O = 1500$ ppm, $p = 5\%$, $R_v = 1$ nm (Fig. 6b), the surface energy change is most significant at intermediate temperatures (1300–1700 K), coinciding with the breakaway swelling regime in nitride fuels. Ronchi *et al.* [7] reported the breakaway swelling critical temperature T_c for mixed nitrides in the interval 1473–1573 K, with an uncertainty of ± 50 K. This range, shown as a shaded band in Fig. 6, overlaps with the temperature of maximum $|\Delta\sigma|$ predicted by our model. The mechanistic modeling by Rizk *et al.* [5] corroborates this picture, attributing the breakaway transition in UN to a change in the gas atom diffusion mechanism occurring around 1500 K. Noting that the range of operating temperatures in liquid metal-cooled fast reactors is 800–2000 K [56], these findings highlight oxygen adsorption as a critical factor in facilitating void nucleation under conditions typical of reactor environments.

The temperature dependence of $|\Delta\sigma|$ is governed by the competition between oxygen adsorption kinetics and bulk solubility. At low temperatures, diffusion is too slow for significant oxygen transport to void surfaces; at high temperatures, the increased bulk solubility of oxygen disfavors surface segregation. As a result, $|\Delta\sigma|$ exhibits a maximum at an intermediate temperature T_{peak} .

A closed-form expression for this peak temperature is derived in Appendix C by analyzing the piecewise monotonicity of $|\Delta\sigma|(T)$. The maximum coincides with the kinetic saturation temperature T^* at which oxygen diffusion first reaches the mean void spacing within the nucleation time, i.e., $\sqrt{D_1(T^*)t} = \lambda_v$. Solving this condition yields:

$$T_{\text{peak}} = \frac{E_m + E_f(v_N^{(b)})}{k \ln \left[\frac{f z \lambda^2 \nu t}{6 c_O \lambda_v^2} \right]}, \quad (33)$$

where f is the correlation factor, z the coordination number, λ the jump distance, and ν the attempt frequency for the $\{O_N : v_N\}$ mechanism (defined in Section 2.3), and $\lambda_v = n_v^{-1/3}$ is the mean free path between voids. The full derivation is given in Appendix C. Because R_v , p , c_O , and t enter only through the logarithm, T_{peak} is controlled primarily by the activation energy scale $E_m + E_f(v_N^{(b)})$ and is only weakly sensitive to microstructural parameters and oxygen concentration.

The weak dependence of $|\Delta\sigma|$ on porosity is also explained analytically in Appendix C.8. The leading $(1-p)/p$ factor in the geometric prefactor of Eq. (7) is exactly canceled by an opposing $p/(1-p)$ factor arising from the surface nitrogen site fraction (Eq. (28)) that enters the segregation ratio (Eq. (32)). The residual porosity dependence is only $1/[N_N^{(b)}] \approx 1$, since $[N_N^{(s)}] \ll 1$ for the void sizes of interest. Furthermore, at T_{peak} , the kinetic factor is saturated ($\alpha_1 = 1$), so the $p^{1/3}$ dependence of α_1 does not contribute at the peak. Numerical evaluation confirms that the peak $|\Delta\sigma|$ varies by less than 10% across $p = 5\text{--}20\%$ for $R_v = 25$ nm (Table C.8).

The dependence on void size is more significant. Since $\lambda_v \sim R_v/p^{1/3}$, larger voids increase the denominator of Eq. (33), shifting T_{peak} to higher temperatures. This also reduces the peak magnitude of $|\Delta\sigma|$ because the segregation drive is weaker at higher temperatures. For example, increasing R_v from 1 nm to 10 nm shifts T_{peak} upward by approximately 180 K (Figs. 6b and 6d). Note this 180 K upward shift observed from the figures can be reproduced quantitatively using Eq. (33), which validates Eq. (33). Oxygen-induced surface stabilization is therefore most effective for small voids during the early stages of nucleation and growth.

It is of interest to examine the contribution of the different terms to the change in surface energy, where $\Delta\sigma(\text{O}_N^{(s)})$ (Eq. (34)) represents the contribution of $\text{O}_N^{(s)}$ to $\Delta\sigma$, with analogous definitions for $\Delta\sigma(\text{O}_i^{(s)})$ (Eq. (35)) and $\Delta\sigma(v_N^{(s)})$ (Eq. (36)).

$$\Delta\sigma(\text{O}_N^{(s)}) = \frac{8}{3} \frac{1-p}{p} \frac{R_v}{a^3} c_O \alpha_1 \frac{[\text{O}_N^{(s)}]}{[\text{O}_N^{(b)}]} E_{\text{ad}}(\text{O}_N^{(s)}). \quad (34)$$

$$\Delta\sigma(\text{O}_i^{(s)}) = \frac{8}{3} \frac{1-p}{p} \frac{R_v}{a^3} c_O \alpha_2 \frac{[\text{O}_i^{(s)}]}{[\text{O}_N^{(b)}]} E_{\text{ad}}(\text{O}_i^{(s)}). \quad (35)$$

$$\Delta\sigma(v_N^{(s)}) = \frac{2}{a^2} [v_N^{(s)}] E_f(v_N^{(s)}). \quad (36)$$

These terms are plotted in Fig. 7. Comparing Figs. 6 and 7a, it can be seen that the adsorption of oxygen as $\text{O}_N^{(s)}$ dominates the $\Delta\sigma$ profile. This is expected since $\text{O}_N^{(s)}$ is the most dominant oxygen adsorption site (refer to Fig. 5c) and it leads to a reduction of surface energy at all temperatures (also see Fig. 3a). From Fig. 7b, it can be observed that $\text{O}_i^{(s)}$ is responsible for the slight surface energy increase in the range of 1020–1520 K in the N-rich profile of Fig. 6. Finally, in Fig. 7c, it is obvious that the formation of N vacancies has a negligible contribution to the surface energy change of voids.

Given that the effect of nitrogen vacancies on surface energy reduction is negligible, and the contribution of $\text{O}_i^{(s)}$ is second order, the $\Delta\sigma$ formula (Eq. (7)) simplifies to a product of four terms: a geometric factor $\eta = ((1-p)/p) \cdot R_v/3$, the oxygen number density $n_O = c_O/\Omega$, a kinetic correction factor α_1 , and an effective segregation energy $E_{\text{seg}}^{\text{eff}} = ([\text{O}_N^{(s)}]/[\text{O}_N^{(b)}]) E_{\text{ad}}(\text{O}_N^{(s)})$. Specifically, we write:

$$\Delta\sigma \approx \left(\frac{(1-p)}{p} \frac{R_v}{3} \right) \cdot \frac{c_O}{\Omega} \cdot \alpha_1 \cdot \left(\frac{[\text{O}_N^{(s)}]}{[\text{O}_N^{(b)}]} E_{\text{ad}}(\text{O}_N^{(s)}) \right) = \eta n_O \alpha_1 E_{\text{seg}}^{\text{eff}}, \quad (37)$$

where c_O is the atomic fraction of oxygen, $\Omega = a^3/8$ is the atomic volume of the matrix (inverse of the atomic number density), p is the porosity, and R_v is the average void radius.

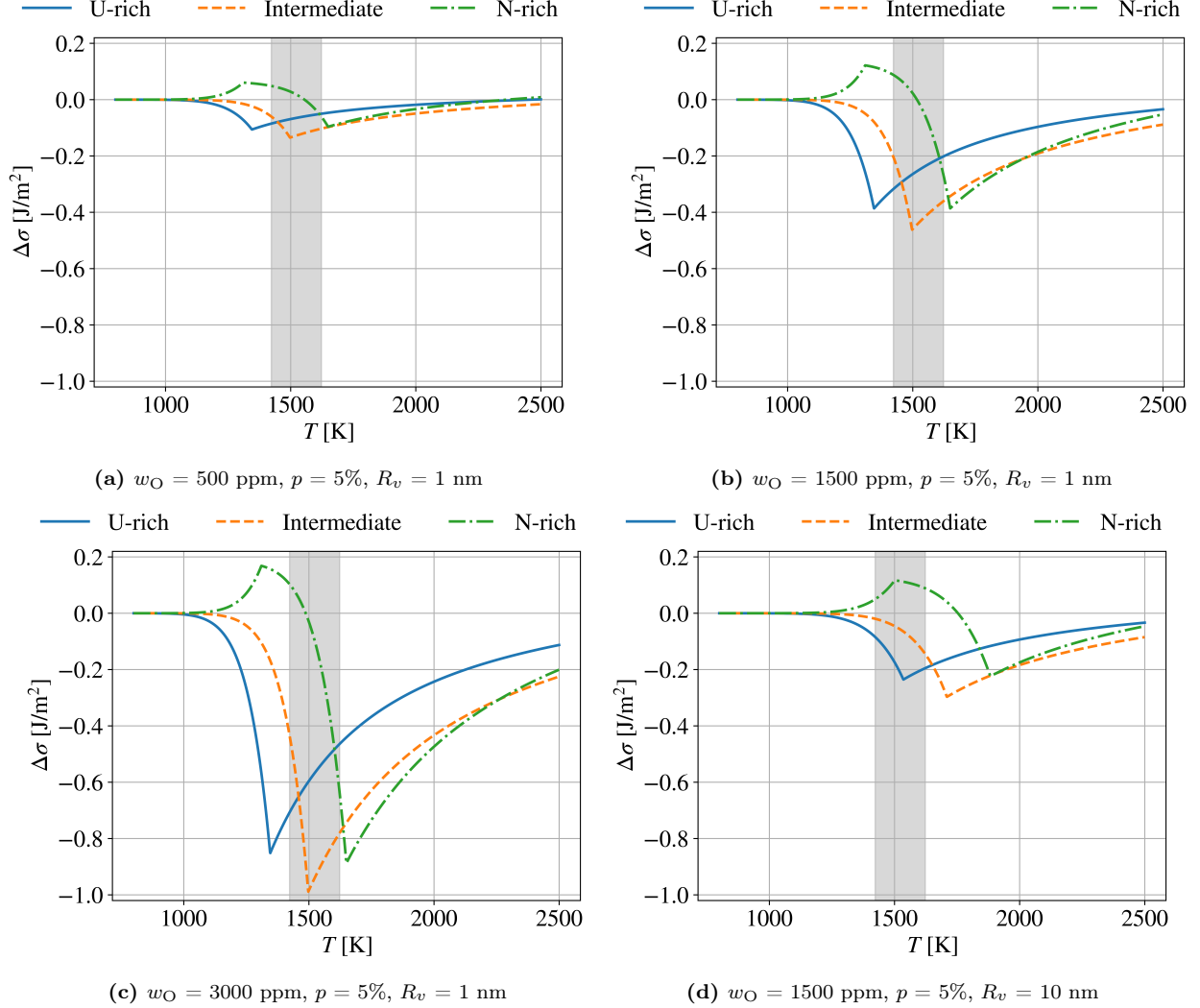


Figure 6: Temperature evolution of surface energy change due to oxygen adsorption and vacancy formation for various porosities, oxygen concentrations, and average void radii. The gray shaded band indicates the breakaway swelling critical temperature range for mixed nitrides (1423–1623 K), based on $T_c = 1473$ –1573 K with ± 50 K uncertainty [7].

The geometric factor η captures the matrix-to-void volume ratio, $(1 - p)/p$, and the void volume-to-surface ratio, $R_v/3$. That is, η represents the average matrix volume per unit void surface area: $(V_{\text{matrix}}/V_{\text{void}}) \times (V_{\text{void}}/A_{\text{void}}) = V_{\text{matrix}}/A_{\text{void}}$, and gives a length scale that quantifies an effective thickness of matrix material supplying each unit of void surface. The kinetic correction factor α_1 accounts for the finite diffusivity of oxygen within the void nucleation timescale (refer to Eq. (8)). Finally, $E_{\text{seg}}^{\text{eff}}$ represents the net energy gain per oxygen atom that segregates to the void surface. This formulation makes explicit that the surface energy reduction is governed by the interplay of thermodynamics, kinetics, and microstructural geometry, and scales linearly with the number of oxygen atoms that successfully reach and adsorb on the available void surface area. Note that the temperature dependence of $|\Delta\sigma|$ is contained in the anti-Arrhenius term $[\text{O}_N^{(s)}]/[\text{O}_N^{(b)}]$ and $\alpha_1 \sim \sqrt{D_1}$.

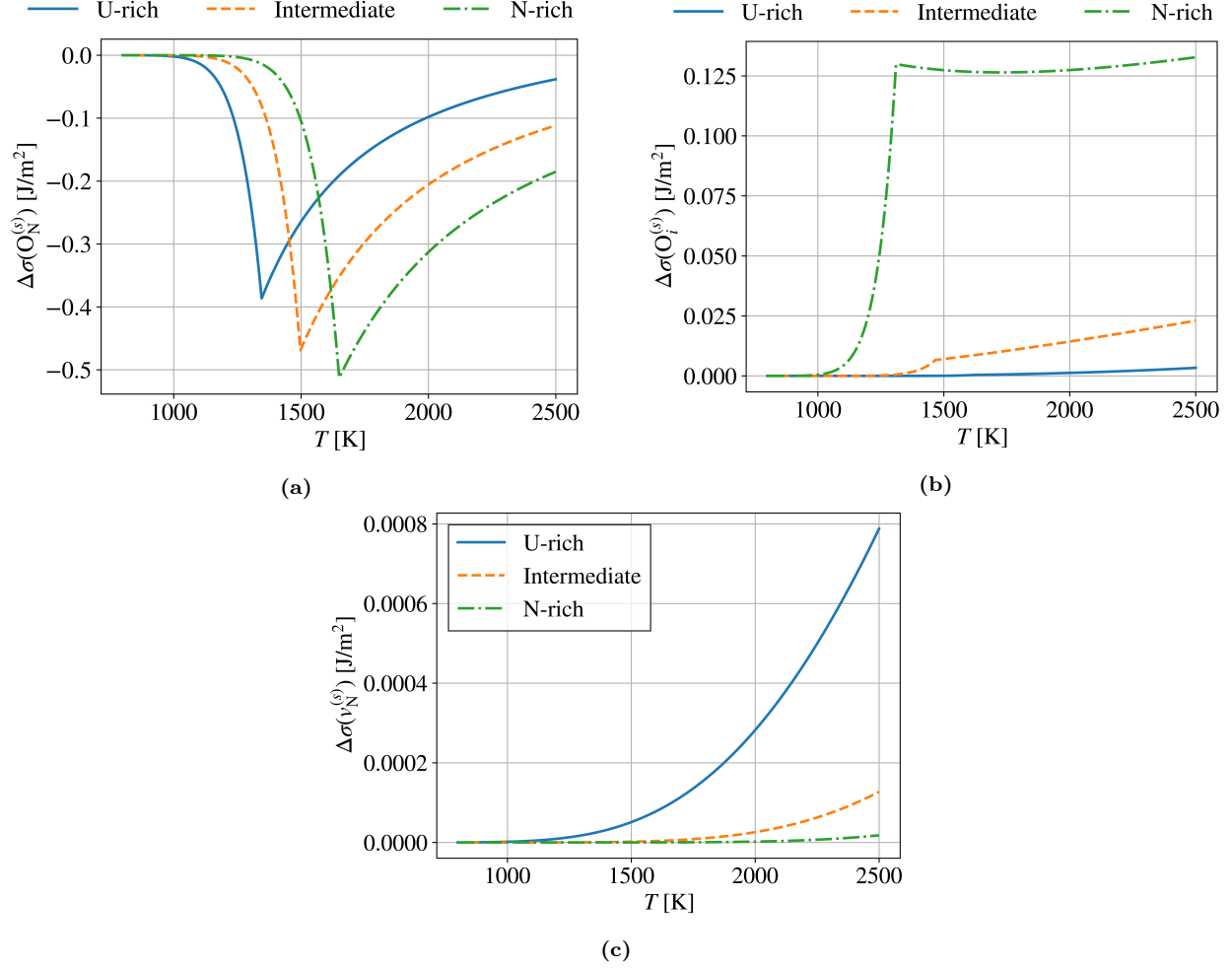


Figure 7: Different terms of the surface energy change due to oxygen adsorption and vacancy formation on void surfaces for $w_{\text{O}} = 1500$ ppm, $p = 5\%$, and $R_v = 1$ nm. (a) $\Delta\sigma(\text{O}_N^{(s)})$, (b) $\Delta\sigma(\text{O}_i^{(s)})$, and (c) $\Delta\sigma(v_N^{(s)})$ are given in Eqs. (34) to (36), respectively.

For the reference case of $w_{\text{O}} = 1500$ ppm, $p = 5\%$, and $R_v = 1$ nm, $|\Delta\sigma|$ is found to be around 0.4 J/m^2 , occurring within the temperature interval 1300–1700 K (see Fig. 6b). To assess the sensitivity of $|\Delta\sigma|$ to key parameters, we vary w_{O} and p while holding R_v fixed at 25 nm, which is a typical size for both dislocations and intergranular bubbles [60]. It has been shown that dislocation bubbles dominate bubble swelling in UN [5–7]. Although the assumptions of our methodology are more representative of bulk voids, we assume that the same surface energy reduction mechanism applies to intragranular (bulk and dislocation) and intergranular bubbles.

Figure 8 represents the maximum $|\Delta\sigma|$, which occurs at intermediate stoichiometric conditions. As explained earlier, because R_v is kept fixed, the temperature at which this maximum occurs is approximately the same for all (w_{O}, p) combinations shown, varying only between ~ 1620 and ~ 1750 K for $R_v = 25$ nm at intermediate stoichiometry. This weak dependence on w_{O} and p follows from Eq. (33), where both parameters enter only through a logarithm. The color map reveals that $|\Delta\sigma|$ is weakly dependent on porosity for $R_v = 25$

nm. The vertical red line marks the oxygen concentration in UN samples from Ronchi *et al.* ($w_{\text{O}} = 1800$ ppm) [6, 7], used by Rizk *et al.* [5] to validate their swelling model. Fission gas bubbles with an average radius of 25 nm are predicted to experience a reduction in surface energy of about 0.3–0.4 J/m², approximately 20% of our calculated surface energy in the oxygen-free case.

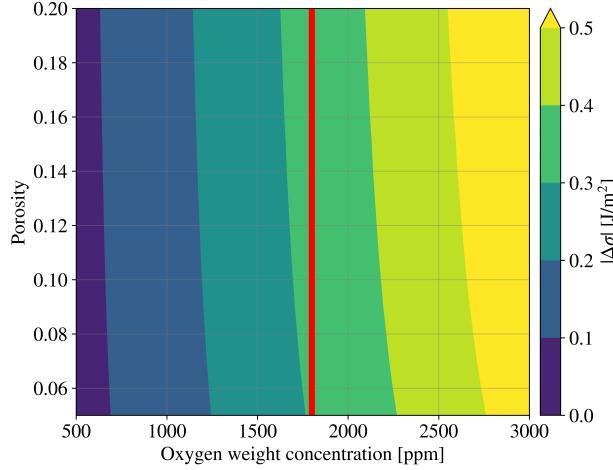


Figure 8: (Color online) Maximum surface energy reduction, $|\Delta\sigma|$ as a function of porosity and oxygen concentration under intermediate stoichiometry for a fixed bubble radius $R_v = 25$ nm, typical of dislocation and grain-boundary bubbles. The red line marks $w_{\text{O}} = 1800$ ppm, the oxygen concentration in UN samples from Ronchi *et al.* [6, 7], used by Rizk *et al.* [5] to validate their swelling model.

In UN samples with an oxygen content of 1600 ppm and carbon content of 300 ppm, Mishchenko *et al.* [27] found 1.5 wt.% UO_2 . Thus, having an oxygen content of, e.g., 1800 ppm will most likely lead to the separation of a UO_2 phase, unless there is some process by which this oxygen is forced to be in solution. There are many solubility limits of oxygen in UN reported in the literature, ranging from 650 ppm to 6640 ppm [61–64]. Lyubimov *et al.* [26] conducted a thermodynamic modeling calculation and found that a carbon concentration of 720 ppm increases the oxygen solubility by a factor of 4–5. Based on this analysis, a carbon content of 720 ppm is large enough to keep an oxygen concentration of 3000 ppm in solution and suppress UO_2 phase separation, thus leading to $|\Delta\sigma| > 0.5$ J/m² (see Fig. 8) and enhanced void swelling in UN.

4. Discussion

This study presents a novel model that integrates first-principles calculations with defect chemistry to quantify the role of oxygen impurities in stabilizing voids and fission gas bubbles in UN. To our knowledge, this is the first model to establish a quantitative link between oxygen concentration and void nucleation thermodynamics in UN. By capturing how oxygen segregates to void surfaces and reduces the surface energy, this work advances the mechanistic understanding of swelling phenomena in oxygen-contaminated nuclear fuels. As shown in Section 3.3, the temperature of maximum $|\Delta\sigma|$ coincides with the experimentally reported breakaway swelling critical temperature for nitride fuels [7] and with the diffusion mechanism

transition identified by Rizk *et al.* [5], reinforcing the connection between oxygen-induced surface stabilization and the onset of accelerated swelling.

The model provides predictive capabilities by mapping the dependence of surface energy reduction on key physical parameters, including temperature, porosity, void radius, and oxygen concentration. Parametric analysis reveals that surface energy reduction is strongly influenced by oxygen concentration and void radius, while being weakly dependent on porosity. In particular, small bulk voids and bubbles experience a significant stabilization effect from oxygen at intermediate temperatures, highlighting the role of oxygen in the early stages of void or fission gas bubble formation. These insights may provide a physically grounded explanation for the association between oxygen content and accelerated swelling observed in experimental studies of mixed uranium carbide/nitride fuels [8]. Furthermore, the generality of the approach allows for its extension to other materials systems and integration into mesoscale models and fuel performance codes, offering a path toward multiscale modeling of chemically assisted swelling in nuclear materials.

Rizk *et al.* adopted a surface energy of 1.11 J/m², computed using the empirical potential developed by Kocevski *et al.* [52]. In contrast, our DFT calculations yield a higher value of 1.59 J/m² for a planar surface. This 0.48 J/m² discrepancy, neglecting curvature corrections, likely stems from the known tendency of the Kocevski potential to underestimate defect energetics in UN [40], a limitation that appears to manifest here as well. Substituting $\sigma = 1.59$ J/m² into the model of Rizk *et al.* would suppress both fission gas swelling and release, necessitating recalibration of other model parameters to preserve agreement with experiment. This discrepancy highlights the need for a mechanism that lowers surface energy in irradiated UN. We propose that oxygen adsorption at bubble surfaces fulfills this role, enabling the levels of swelling and gas release observed experimentally. A reduction in surface energy thus appears essential to reconcile the mechanistic model with experimental data. Notably, the oxygen concentration reported by Ronchi *et al.* ($w_O = 1800$ ppm) corresponds to a predicted surface energy reduction of 0.3–0.4 J/m², closely matching the gap between our DFT value and that used by Rizk *et al.*

To quantify the impact of oxygen-induced surface energy reduction on void nucleation, we invoke classical nucleation theory. The free energy barrier for homogeneous void nucleation is [65]:

$$\Delta G^* = \frac{16\pi\sigma^3}{3(\Delta g_v)^2}, \quad (38)$$

where Δg_v is the volumetric free energy driving force for void formation. Since $\Delta G^* \sim \sigma^3$, the ratio of nucleation barriers with and without oxygen is:

$$\frac{\Delta G_O^*}{\Delta G_{\text{clean}}^*} = \left(\frac{\sigma_{\text{clean}} + \Delta\sigma}{\sigma_{\text{clean}}} \right)^3 = \left(1 + \frac{\Delta\sigma}{\sigma_{\text{clean}}} \right)^3. \quad (39)$$

For $\sigma_{\text{clean}} = 1.59$ J/m² and $\Delta\sigma \approx -0.4$ J/m² (intermediate stoichiometry, $w_O = 1500$ ppm, $R_v = 1$ nm), this gives $\sigma_O \approx 1.19$ J/m² and:

$$\frac{\Delta G_O^*}{\Delta G_{\text{clean}}^*} = \left(\frac{1.19}{1.59} \right)^3 \approx 0.42, \quad (40)$$

corresponding to a $\sim 58\%$ reduction in the nucleation barrier. This substantial lowering of ΔG^* translates into a dramatically increased nucleation rate, $J \sim \exp(-\Delta G^*/kT)$ [65], consistent with the experimentally observed enhancement of void swelling in oxygen-contaminated UN.

Despite its contributions, the present model includes several simplifications that merit further investigation. First, the surface adsorption energy of oxygen is assumed to be independent of surface coverage, which may not hold at high impurity concentrations where impurity-impurity interactions become significant. Additionally, configurational and vibrational entropy contributions are neglected, which may affect the temperature dependence of defect formation and migration energies, particularly at elevated temperatures.

Further, the kinetic parameter used to model oxygen segregation is introduced in an *ad hoc* manner, limiting the model's capability to capture time-dependent or rate-limited processes. To assess sensitivity to this parameter, we note that $\alpha_i \sim \sqrt{t}$ (Eq. (8)), so that $|\Delta\sigma|$ in the kinetically limited regime scales as $\sqrt{t}$. Varying t from 10 to 100 hours, $|\Delta\sigma|$ decreases by $\sim 51\%$ for $t = 10$ hours and increases by $\sim 54\%$ for $t = 100$ hours, relative to the baseline $t = 42$ hours. Note that the temperature at which $|\Delta\sigma|$ is maximized also depends on t but only logarithmically (through Eq. (33)), giving a downward shift of approximately 80 K over the order-of-magnitude range $t = 10$ –100 hours for the reference case of $R_v = 1$ nm, $w_O = 1500$ ppm, and $p = 0.05$.

Also, the influence of sinks such as grain boundaries and second-phase precipitates, which may trap oxygen or alter defect transport, is also not incorporated.

Another limitation of the present model is that all surface properties are derived from planar (001) slab calculations, whereas real voids may expose higher-index facets such as $\{110\}$ and $\{111\}$, and exhibit curvature effects at small radii. Regarding curvature, if the total energy of a spherical void is written as [66–68]:

$$E_v = 4\pi R_v^2 \sigma \left(1 + \frac{\delta}{R_v} \right) \quad (41)$$

where δ is a Tolman-like length on the order of ~ 1 nm, the correction to the absolute surface energy (equal to δ/R_v) is substantial for the smallest voids considered here: 100% at $R_v = 1$ nm, 50% at $R_v = 2$ nm, diminishing to 10% at $R_v = 10$ nm and 4% at $R_v = 25$ nm. However, the central quantity predicted by our model is not σ itself but the change in surface energy, $\Delta\sigma$, due to oxygen adsorption. As demonstrated in the derivation of Eq. (7), the pristine surface energy cancels identically: $\Delta\sigma$ depends only on adsorption energies (E_{ad}), defect formation energies (E_f), defect concentrations, and kinetic correction factors, none of which are functions of σ . Curvature corrections to the absolute surface energy therefore do not propagate into the predicted $\Delta\sigma$, and the conclusions regarding oxygen-induced surface energy reduction remain quantitatively unchanged irrespective of the curvature model adopted.

Regarding faceting, the (001) surface is the most stable low-index surface of UN [18] and is expected to dominate the equilibrium shape of voids in rocksalt-structured compounds. While $\{110\}$ and $\{111\}$ facets may be present, particularly in non-equilibrium or irradiation-produced voids, their contribution to the total void surface area is expected to be secondary. In principle, the oxygen adsorption energies on these higher-index surfaces may differ from

those on the (001) surface, which would introduce facet-dependent corrections to $\Delta\sigma$. Quantifying these corrections would require dedicated slab calculations for each facet orientation, which is beyond the scope of the present work. Nevertheless, because the dominant contribution to $\Delta\sigma$ arises from the segregation of oxygen into surface nitrogen vacancies ($\text{O}_\text{N}^{(\text{s})}$), which is governed primarily by the local coordination environment of the nitrogen sublattice rather than the global surface orientation, we expect the qualitative trends reported here to be robust across facets. A systematic investigation of facet-dependent adsorption energies and their impact on void stabilization is a natural extension of this work.

A further limitation is that the current model isolates oxygen as the sole impurity species. In reality, fission products and other impurities, such as carbon, krypton, and xenon, may co-segregate or interact with oxygen, altering the energetics and morphology of vacancy clusters. Notably, our calculations indicate that substitutional oxygen on nitrogen sites binds to uranium-site vacancies and noble gas atoms (i.e., Kr_U , Xe_U) with energies around -0.4 eV. Whether this binding promotes the stabilization of voids or instead alters gas mobility and clustering behavior is not yet fully understood. Future studies should assess how such interactions affect oxygen-assisted swelling in fission environments.

Finally, while the role of carbon in increasing oxygen solubility and suppressing UO_2 precipitation in UN is qualitatively known, the impact of carbon content on void stabilization was not explicitly addressed in this model. Given that carbon can modulate the chemical potential of oxygen and influence its incorporation and redistribution, a natural extension of this work is to explore the combined effects of carbon and oxygen on defect energetics and swelling behavior in UN. The framework developed here is directly applicable to this scenario.

5. Conclusions

This work develops a first-principles-based thermodynamic framework that quantifies how oxygen impurities interact with void surfaces in UN, leading to a reduction in the surface energy, $|\Delta\sigma|$. The model incorporates DFT-derived energetics, oxygen diffusivity, defect chemistry, and microstructural parameters to assess the extent of surface stabilization.

Key findings can be summarized as follows. Substitutional oxygen on nitrogen surface sites ($\text{O}_\text{N}^{(\text{s})}$) is the dominant contributor to surface energy reduction. Oxygen in surface hollow sites ($\text{O}_\text{i}^{(\text{s})}$) has a minor, and occasionally adverse, effect on $|\Delta\sigma|$. The dependence of $|\Delta\sigma|$ on key parameters is summarized below:

- *Oxygen concentration (w_O):* The surface energy reduction scales linearly with oxygen content.
- *Void radius (R_v):* $|\Delta\sigma|$ is most pronounced for small voids. Larger voids require higher temperatures to reach meaningful oxygen coverage and experience a slightly smaller $|\Delta\sigma|$.
- *Porosity (p):* The surface energy reduction is found to be weakly dependent on porosity.

For small voids and bubbles ($R_\text{v} = 1\text{--}10$ nm), $|\Delta\sigma|$ exhibits three regimes in its dependence on temperature:

- *Low temperatures:* $|\Delta\sigma| \approx 0$ due to limited oxygen diffusivity.

- *Intermediate temperatures:* $|\Delta\sigma|$ reaches its maximum as the competing effects of oxygen solubility and diffusion kinetics are optimally balanced. This regime coincides with the onset of breakaway swelling in UN.
- *High temperatures:* $|\Delta\sigma|$ decreases due to enhanced bulk solubility, which disfavors surface segregation.

The reduction in surface energy can be approximately expressed as $|\Delta\sigma| \approx \eta n_O \alpha_1 |E_{\text{seg}}^{\text{eff}}|$ (Eq. (37)), where η is a geometric factor that depends on p and R_v , n_O is the oxygen number density, α_1 is a kinetic correction factor, and $E_{\text{seg}}^{\text{eff}}$ is the effective segregation energy. The peak temperature is given by Eq. (33). The reduction in σ lowers the thermodynamic barrier for void and bubble formation. For the reference case of $w_O = 1800$ ppm in UN [6], our model predicts a surface energy reduction of about 0.3–0.4 J/m² for dislocation and grain-boundary bubbles. This aligns closely with the discrepancy between our DFT-calculated surface energy (1.59 J/m²) and the value adopted by Rizk *et al.* [5] in their swelling model (1.11 J/m²). The oxygen-induced reduction in σ thus can offer a mechanistic explanation for the experimentally observed levels of swelling and fission gas release in UN.

Overall, this work provides a physically rigorous model for incorporating oxygen effects into mechanistic swelling models of UN. The framework is extendable to other chemical impurities such as carbon and can be integrated into mesoscale and fuel performance models to predict chemically-assisted bubble swelling in nuclear fuels.

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Appendix A. Derivation of the formulas for $[N_N^{(s)}]$ and $[v_i^{(s)}]$

In the following, a denotes the UN lattice constant, p the porosity (void volume fraction), R_v the average void radius, and V_0 the total system volume. $[N_N^{(s)}]$ is the site fraction of nitrogen atoms on void surfaces, and $[v_i^{(s)}]$ is the site fraction of vacant interstitial sites on void surfaces, both expressed per formula unit.

The area associated with an N atom on the surface is $a^2/2$, where a is the lattice constant. The total surface area of the internal voids is $N_v \times 4\pi R_v^2$, where N_v is the total number of

voids, calculated as:

$$N_v = \frac{pV_0}{\frac{4}{3}\pi R_v^3}, \quad (\text{A.1})$$

where V_0 is the total system volume. Then, the number of $N_N^{(s)}$ sites is:

$$\text{Number of } N_N^{(s)} = \frac{N_v \times 4\pi R_v^2}{a^2/2}. \quad (\text{A.2})$$

To get the site fraction of $N_N^{(s)}$, we need to divide its total number by the number of formula units. The number of formula units is:

$$\text{Number of formula units} = \frac{V_0(1-p)}{a^3/4}, \quad (\text{A.3})$$

where $V_0(1-p)$ is the occupied volume, and $a^3/4$ is the volume of a single formula unit. Dividing Eq. (A.2) by Eq. (A.3), we arrive at:

$$[N_N^{(s)}] = \frac{pV_0}{\frac{4}{3}\pi R_v^3} \frac{4\pi R_v^2}{a^2/2} \frac{a^3/4}{V_0(1-p)} = \frac{3}{2} \frac{p}{1-p} \frac{a}{R_v}. \quad (\text{A.4})$$

The same steps can be followed to get an expression for $[v_i^{(s)}]$, provided that we pay attention to the following differences: First, the area occupied by a vacant site on the surface is $a^2/4$. Then, the number of $v_i^{(s)}$ sites is:

$$\text{Number of } v_i^{(s)} = \frac{N_v \times 4\pi R_v^2}{a^2/4}. \quad (\text{A.5})$$

To get the site fraction of $v_i^{(s)}$, we need to divide its total number by the number of formula units in the occupied volume (i.e., Eq. (A.3)), and then divide by 2 because there are 2 tetrahedral interstitial sites per formula unit, either in the bulk material or on the surface. In the case of $[N_N^{(s)}]$, there was one N atom (or site) per formula unit. Then, the site fraction of $v_i^{(s)}$ is:

$$[v_i^{(s)}] = \frac{1}{2} \frac{pV_0}{\frac{4}{3}\pi R_v^3} \frac{4\pi R_v^2}{a^2/4} \frac{a^3/4}{V_0(1-p)} = \frac{3}{2} \frac{p}{1-p} \frac{a}{R_v}, \quad (\text{A.6})$$

which is the same expression for $[N_N^{(s)}]$. Note that both Eqs. (A.4) and (A.6) are independent of V_0 . As outlined in Section 3.1, the adsorption energy of O atoms in above-U sites is the same as that of O in a hollow site (Table 6). To account for this in our model, the available adsorption sites are doubled by multiplying $[v_i^{(s)}]$ by 2. In this way, the analysis is simplified by only considering $O_i^{(s)}$. The final expression reads:

$$[v_i^{(s)}] = \frac{3p}{1-p} \frac{a}{R_v}. \quad (\text{A.7})$$

Appendix B. Derivation of the formula for surface energy change as a function of adsorbate atom concentration

Here, σ_0 is the pristine surface energy, E_{ad} the adsorption energy, E_f the defect formation energy, N_{ad} the number of adsorbed atoms, A the total surface area, and μ_{ad} the chemical potential of the adsorbate. $[\text{O}_{\text{N}}^{(\text{s})}]$ and $[\text{O}_{\text{i}}^{(\text{s})}]$ denote the site fractions of substitutional and interstitial oxygen on void surfaces, respectively, while $[\text{O}_{\text{N}}^{(\text{b})}]$ is the site fraction of substitutional oxygen in the bulk.

Suppose we have an impurity atom that got adsorbed on a pristine surface. The surface energy after absorption is:

$$\sigma = \sigma_0 + \Delta\sigma, \quad (\text{B.1})$$

where σ_0 is the energy of the pristine surface before adsorption, and $\Delta\sigma$ is the surface energy change due to impurity adsorption. The change in the surface energy can be expressed as:

$$\Delta\sigma = \sigma_{\text{surf+ad}} - \sigma_{\text{surf}}, \quad (\text{B.2})$$

where:

$$\sigma_{\text{surf}} = \frac{1}{A} [E_{\text{surf}} - N_{\text{U}}\mu_{\text{U}} - N_{\text{N}}\mu_{\text{N}}], \quad (\text{B.3})$$

and:

$$\sigma_{\text{surf+ad}} = \frac{1}{A} [E_{\text{surf+ad}} - N_{\text{U}}\mu_{\text{U}} - N_{\text{N}}\mu_{\text{N}} - N_{\text{ad}}\mu_{\text{ad}}]. \quad (\text{B.4})$$

Then,

$$\Delta\sigma = \frac{1}{A} [E_{\text{surf+ad}} - E_{\text{surf}} - N_{\text{ad}}\mu_{\text{ad}}]. \quad (\text{B.5})$$

From the definition of the adsorption energy, i.e., [Eq. \(6\)](#), the change in the surface energy is:

$$\Delta\sigma = \frac{1}{A} [N_{\text{ad}}(E_{\text{ad}} + \mu_{\text{ad}}) - N_{\text{ad}}\mu_{\text{ad}}] = \frac{1}{A} N_{\text{ad}} E_{\text{ad}}. \quad (\text{B.6})$$

If we have a specific concentration of an impurity that can get adsorbed on different sites on the surface, with each site type having its own adsorption energy, then:

$$\Delta\sigma = \frac{1}{A} \left[\sum_i N_{\text{ad}}^i E_{\text{ad}}^i \right]. \quad (\text{B.7})$$

The same line of thought can be used to derive an expression for the surface energy change upon introducing vacancies to a previously pristine surface. For the general case where we introduce both adsorbate atoms and vacancies to a surface, the total surface energy change is:

$$\Delta\sigma = \frac{1}{A} \left[\sum_i N_{\text{ad}}^i E_{\text{ad}}^i + \sum_j N_{\text{d}}^j E_{\text{f}}^j \right]. \quad (\text{B.8})$$

This can be related to the relative concentrations calculated earlier by noticing that:

$$N_{\text{ad}}^i = N_{\text{O}} \frac{N_{\text{ad}}^i}{N_{\text{O}}} \approx N_{\text{O}} \frac{[\text{O}_{\text{X}}^{(\text{s})}]}{[\text{O}_{\text{N}}^{(\text{b})}]}, \quad (\text{B.9})$$

694 where N_O is the total number of oxygen atoms in the sample and $X \in \{N, i\}$. Finally, we can
 695 relate N_d , i.e., the total number of surface defects, to the concentrations calculated earlier
 696 by a similar argument. Assuming the surface defect is $v_N^{(s)}$, we have:

$$N_d = [v_N^{(s)}] \times \text{Number of } N_N^{(s)} = [v_N^{(s)}] \frac{A_v}{a^2/2}, \quad (\text{B.10})$$

697 where we have made use of Eq. (A.2) with A_v being the total area of the void surface.

698 The total number of oxygen atoms in the system is not a useful parameter. Instead, we
 699 need the formulas to be expressed in terms of some measure of oxygen concentration. The
 700 number of voids is expressed in terms of both the void surface area and the void volume as:

$$N_v = \frac{A_v}{4\pi R_v^2} = \frac{pV_0}{\frac{4}{3}\pi R_v^3}. \quad (\text{B.11})$$

701 From Eq. (B.11), the total void surface area is:

$$A_v = \frac{3pV_0}{R_v} = \frac{3}{R_v} \frac{p}{1-p} V. \quad (\text{B.12})$$

702 Then,

$$\frac{N_O}{A_v} = \frac{R_v}{3} \frac{1-p}{p} \frac{N_O}{V} = \frac{R_v}{3} \frac{1-p}{p} \frac{N_O}{2N_{UN} \cdot a^3/8} \approx \frac{8}{3} \frac{1-p}{p} \frac{R_v}{a^3} c_O, \quad (\text{B.13})$$

703 where N_{UN} is the number of formula units in the occupied volume, $a^3/8$ is the average
 704 volume per atom of UN, and c_O is the atomic fraction of O atoms which is approximately
 705 $c_O \approx N_O/(2N_{UN})$. The oxygen impurity concentration in UN is usually expressed as parts
 706 per million (ppm) or weight fractions. The relation between atomic fraction, c_O , and weight
 707 fraction, w_O , is:

$$c_O = \frac{M_{UN}w_O}{M_{UN}w_O + 2M_O(1-w_O)}, \quad (\text{B.14})$$

708 where M_{UN} and M_O are the molar weights of UN and O, respectively. From Eq. (B.14), w_O
 709 can be expressed in terms of c_O as:

$$w_O = \frac{2M_Oc_O}{2M_Oc_O + M_{UN}(1-c_O)}. \quad (\text{B.15})$$

710 Substituting Eqs. (B.9), (B.10) and (B.13) into Eq. (B.8), the change in surface energy
 711 becomes:

$$\Delta\sigma = \frac{8}{3} \frac{1-p}{p} \frac{R_v}{a^3} c_O \left[\frac{[O_N^{(s)}]}{[O_N^{(b)}]} E_{ad}(O_N^{(s)}) + \frac{[O_i^{(s)}]}{[O_N^{(b)}]} E_{ad}(O_i^{(s)}) \right] + \frac{2}{a^2} [v_N^{(s)}] E_f(v_N^{(s)}). \quad (\text{B.16})$$

712 To take kinetics into account, the concentrations of the different oxygen surface defects
 713 must be corrected for the low oxygen diffusivity at low temperatures. Note that the increased
 714 oxygen solubility is inherently incorporated in the thermodynamic model. According to
 715 Zinkle and Lee [12], as a first approximation, the concentrations of oxygen surface defects
 716 can be multiplied by a parameter α that we term the kinetic correction, and that depends

on the diffusivity of different oxygen defect types. Thus, the final expression for the change in surface energy becomes:

$$\Delta\sigma = \frac{8}{3} \frac{1-p}{p} \frac{R_v}{a^3} c_O \left[\alpha_1 \frac{[\text{O}_N^{(s)}]}{[\text{O}_N^{(b)}]} E_{\text{ad}}(\text{O}_N^{(s)}) + \alpha_2 \frac{[\text{O}_i^{(s)}]}{[\text{O}_N^{(b)}]} E_{\text{ad}}(\text{O}_i^{(s)}) \right] + \frac{2}{a^2} [v_N^{(s)}] E_f(v_N^{(s)}), \quad (\text{B.17})$$

where α_1 and α_2 are the kinetic corrections for the defects $\text{O}_N^{(s)}$ and $\text{O}_i^{(s)}$, respectively. Conceptually, α_i stems from the fraction of diffusing oxygen atoms that encounter a void embryo during the void nucleation time, and, thus, can be defined as [12]:

$$\alpha_i = \begin{cases} \sqrt{D_i t}/\lambda_v & \text{if } \sqrt{D_i t}/\lambda_v < 1, \\ 1 & \text{if } \sqrt{D_i t}/\lambda_v \geq 1, \end{cases} \quad (\text{B.18})$$

where D_i is the diffusivity of the relevant defect, t is the void nucleation time, and λ_v is the mean free path between voids, defined as $\lambda_v = n_v^{-1/3}$ [69], where $n_v = p/(\frac{4}{3}\pi R_v^3)$ is the void number density.

Appendix C. Closed-form expression for the temperature of maximum $|\Delta\sigma|$

This appendix derives a closed-form expression for the temperature at which $|\Delta\sigma|$ is maximized. The derivation proceeds in four steps: we first state the assumptions, then factor the reduced $|\Delta\sigma|(T)$ into terms with known monotonicity, establish piecewise monotonicity on either side of a kinetic saturation temperature T^* , and conclude that $T_{\text{peak}} = T^*$. The closed-form result for T^* is given in Eq. (C.14), and its implications for the weak porosity dependence of $|\Delta\sigma|$ are discussed in Appendix C.8.

Appendix C.1. Assumptions

The derivation rests on the following assumptions, each consistent with the modeling choices in Section 3.3:

- (i) *Dominant channel.* Only the $\text{O}_N^{(s)}$ term in Eq. (7) is retained; the $\text{O}_i^{(s)}$ and $v_N^{(s)}$ contributions are neglected. This is justified by Fig. 7, where $\Delta\sigma(\text{O}_N^{(s)})$ dominates by at least an order of magnitude.
- (ii) *Fixed stoichiometry.* The vacancy formation energies $E_f(v_N^{(b)})$ and $E_f(v_N^{(s)})$ are held at their values for a given stoichiometric condition.
- (iii) *Dilute defects.* Concentrations follow Arrhenius statistics with unit prefactor, e.g., $[v_N^{(b)}] = \exp[-E_f(v_N^{(b)})/(kT)]$ and $[\text{O}_N^{(b)}] \approx c_O$.
- (iv) *Single transport channel.* The kinetic correction α_1 is computed from the $\{\text{O}_N : v_N\}$ diffusivity (Section 3.3).
- (v) *Existence of T^* .* The argument of the logarithm in Eq. (C.14) must exceed unity, which is verified for all parameter combinations considered in this work.

Appendix C.2. Factorized form

Retaining only the dominant $O_N^{(s)}$ term, Eq. (7) takes the form:

$$|\Delta\sigma|(T) = \underbrace{\frac{8}{3} \frac{1-p}{p} \frac{R_v}{a^3} c_O}_{\mathcal{P}} \alpha_1(T) g(T) |E_{ad}(T)|, \quad (C.1)$$

where $\mathcal{P}$ is a T -independent prefactor. Three factors carry temperature dependence: the kinetic correction α_1 , the segregation ratio $g \equiv [O_N^{(s)}]/[O_N^{(b)}]$, and the adsorption energy E_{ad} evaluated with respect to the impurity chemical potential (Section 2.2). We now derive the explicit T -dependence of each.

Kinetic factor. The kinetic correction $\alpha_1(T) = \min[\sqrt{D_1(T)t}/\lambda_v, 1]$ (Eq. (8)) is continuous but non-differentiable at the *saturation temperature* T^* defined by $\sqrt{D_1(T^*)t} = \lambda_v$. The effective oxygen diffusivity via the $\{O_N : v_N\}$ mechanism is:

$$D_1(T) = f \frac{[v_N^{(b)}]}{c_O} D_{\{O_N:v_N\}} = \frac{f z \lambda^2 \nu}{6 c_O} \exp\left[-\frac{E_m + E_f(v_N^{(b)})}{kT}\right], \quad (C.2)$$

where the second equality uses $D_d = \frac{1}{6} z \lambda^2 \nu \exp[-E_m/(kT)]$ and $[v_N^{(b)}] = \exp[-E_f(v_N^{(b)})/(kT)]$. In the unsaturated regime ($T < T^*$):

$$\alpha_1(T) = C_\alpha \exp\left[-\frac{E_m + E_f(v_N^{(b)})}{2kT}\right], \quad C_\alpha \equiv \frac{1}{\lambda_v} \sqrt{\frac{f z \lambda^2 \nu t}{6 c_O}}. \quad (C.3)$$

Segregation factor. Combining Eq. (32) with the K_5 equilibrium (Eq. (27)):

$$g(T) = \frac{[N_N^{(s)}]}{[N_N^{(b)}]} \exp\left[+\frac{E_{seg}}{kT}\right], \quad (C.4)$$

$$E_{seg} \equiv -[E_f(v_N^{(s)}) - E_f(v_N^{(b)}) + E_{ad}(O_N^{(s)}) - E_{inc}(O_N^{(b)})], \quad (C.5)$$

where both E_{ad} and E_{inc} in the definition of E_{seg} contain μ_O , which cancels in their difference. That is, E_{seg} is independent of the choice of oxygen reference state and is constructed entirely from raw DFT total-energy differences. Note that the prefactor $[N_N^{(s)}]/[N_N^{(b)}]$ is T -independent.

Substituting the DFT values from Tables 1, 2 and 6:

$$E_{seg} = -[(-0.76) + (0.43)] = 0.33 \text{ eV}, \quad (C.6)$$

where the segregation energy of v_N is $E_f(v_N^{(s)}) - E_f(v_N^{(b)}) = -0.76 \text{ eV}$ (Table 1) and $E_{ad}(O_N^{(s)}) - E_{inc}(O_N^{(b)}) = -6.80 - (-7.23) = 0.43 \text{ eV}$ (Tables 2 and 6). Both differences are stoichiometry-independent, so $E_{seg} = 0.33 \text{ eV}$ for all conditions.

766 *Adsorption-energy factor.* From the definition of the adsorption energy (Eq. (6)), the ad-
 767 sorption energy referred to the impurity chemical potential is:

$$E_{\text{ad}}(T) = \frac{1}{N_{\text{ad}}} (E_{\text{surf+ad}} - E_{\text{surf}}) - \mu_{\text{O}}(T), \quad (\text{C.7})$$

768 where $\frac{1}{N_{\text{ad}}} (E_{\text{surf+ad}} - E_{\text{surf}})$ is the raw (reference-free) adsorption energy, a T -independent
 769 quantity obtained directly from DFT slab calculations. From Eq. (16), using $[\text{O}_{\text{N}}^{(\text{b})}] \approx c_{\text{O}}$
 770 and $[v_{\text{N}}^{(\text{b})}] = \exp[-E_f(v_{\text{N}}^{(\text{b})})/(kT)]$:

$$\mu_{\text{O}}(T) = E(\text{O}_{\text{N}}) - E(v_{\text{N}}) + E_f(v_{\text{N}}^{(\text{b})}) + kT \ln c_{\text{O}}. \quad (\text{C.8})$$

771 Substituting:

$$\begin{aligned} E_{\text{ad}}(T) &= \frac{1}{N_{\text{ad}}} (E_{\text{surf+ad}} - E_{\text{surf}}) - E(\text{O}_{\text{N}}) + E(v_{\text{N}}) - E_f(v_{\text{N}}^{(\text{b})}) - kT \ln c_{\text{O}}. \\ &\quad \underbrace{\hspace{10em}}_{\equiv E_{\text{ad}}^{(0)}, T\text{-independent}} \end{aligned} \quad (\text{C.9})$$

772 Note that $E_{\text{ad}}^{(0)}$ is built entirely from raw DFT total energies and the vacancy formation
 773 energy; no oxygen chemical potential appears. Since $E_{\text{ad}} < 0$ (favorable adsorption) and
 774 $\ln c_{\text{O}} < 0$ (dilute impurity), the magnitude $|E_{\text{ad}}(T)| = -E_{\text{ad}}^{(0)} + kT \ln c_{\text{O}}$ is strictly *decreasing*
 775 in T (also see Fig. 3), with $d|E_{\text{ad}}|/dT = k \ln c_{\text{O}} < 0$.

776 Appendix C.3. Piecewise monotonicity and peak location

777 The function $|\Delta\sigma|(T)$ is continuous but not differentiable at $T = T^*$. We establish its
 778 monotonicity separately on each side.

779 Appendix C.3.1. Unsaturated regime ($T < T^*$)

780 Substituting Eqs. (C.3), (C.5) and (C.9) into Eq. (C.1) and taking the logarithmic deriva-
 781 tive:

$$\frac{d}{dT} \ln |\Delta\sigma| = \underbrace{\frac{1}{kT^2} \left[\frac{E_m + E_f(v_{\text{N}}^{(\text{b})})}{2} - E_{\text{seg}} \right]}_{> 0 \text{ if Eq. (C.11) holds}} + \underbrace{\frac{k \ln c_{\text{O}}}{|E_{\text{ad}}(T)|}}_{< 0}. \quad (\text{C.10})$$

782 The first term is positive provided:

$$E_m + E_f(v_{\text{N}}^{(\text{b})}) > 2 E_{\text{seg}}. \quad (\text{C.11})$$

783 The second term is negative since $\ln c_{\text{O}} < 0$. The sign of the sum determines whether $|\Delta\sigma|$
 784 is increasing or decreasing. The inequality Eq. (C.11) is verified in Appendix C.4, where
 785 it is shown to be satisfied with a margin of $\sim 2\text{--}2.5$ eV for all stoichiometric conditions.
 786 Furthermore, a numerical evaluation over the full physically relevant parameter domain
 787 ($T \in [800, 2500]$ K, $w_{\text{O}} \in [500, 3000]$ ppm, all three stoichiometries) confirms that the positive
 788 first term dominates the negative second term, so the sum in Eq. (C.10) is strictly positive
 789 throughout the unsaturated regime.

	U-rich	Intermediate	N-rich
$E_m + E_f(v_N^{(b)})$ [eV]	4.70	5.19	5.68
$2E_{\text{seg}}$ [eV]	0.66	0.66	0.66
$\frac{1}{2}[E_m + E_f(v_N^{(b)})] - E_{\text{seg}}$ [eV]	2.02	2.27	2.51

Appendix C.3.2. Saturated regime ($T > T^*$)

In this regime $\alpha_1 = 1$, so:

$$\frac{d}{dT} \ln |\Delta\sigma| = -\frac{E_{\text{seg}}}{kT^2} + \frac{k \ln c_O}{|E_{\text{ad}}(T)|}. \quad (\text{C.12})$$

Both terms are negative ($E_{\text{seg}} > 0$ and $\ln c_O < 0$), so $|\Delta\sigma|$ is *strictly decreasing* for $T > T^*$, unconditionally. This conclusion is independent of the kinetic model.

Appendix C.3.3. Peak location

Since $|\Delta\sigma|$ is strictly increasing for $T < T^*$ and strictly decreasing for $T > T^*$, continuity implies:

$$T_{\text{peak}} = T^*. \quad (\text{C.13})$$

Because T^* is a kink (non-differentiable point) of $|\Delta\sigma|(T)$, this result is obtained via piecewise monotonicity rather than by setting a derivative to zero.

Appendix C.4. Verification of the monotonicity condition

The inequality Eq. (C.11) is evaluated for each stoichiometric condition using the DFT values in Table 1 and the migration energy $E_m = 3.06$ eV: The margin of ~ 2 – 2.5 eV in the first term of Eq. (C.10) ensures strong separation from the competing negative term. The negative contribution is bounded in magnitude by $k|\ln c_O|/|E_{\text{ad}}|$. For $c_O \sim 10^{-3}$ – 10^{-2} and $|E_{\text{ad}}| \gtrsim 1$ eV, this term is $\lesssim 6 \times 10^{-4} \text{ K}^{-1}$. In contrast, the first term remains $\gtrsim 10^{-3} \text{ K}^{-1}$ over $T = 800$ – 2500 K, and thus dominates throughout the range of interest.

Appendix C.5. Closed-form expression for T^*

Solving $\sqrt{D_1(T^*)}t = \lambda_v$ using Eq. (C.2):

$$T_{\text{peak}} = T^* = \frac{E_m + E_f(v_N^{(b)})}{k \ln \left[\frac{f z \lambda^2 \nu t}{6 c_O \lambda_v^2} \right]}. \quad (\text{C.14})$$

Substituting $\lambda_v = (4\pi R_v^3/3p)^{1/3}$:

$$T_{\text{peak}} = \frac{E_m + E_f(v_N^{(b)})}{k \ln \left[\frac{f z \lambda^2 \nu t}{6 c_O} \left(\frac{3p}{4\pi R_v^3} \right)^{2/3} \right]}. \quad (\text{C.15})$$

The formula is meaningful only when the argument of the logarithm exceeds unity, i.e., when $f z \lambda^2 \nu t > 6 c_O \lambda_v^2$. Physically, this means oxygen must be capable of diffusing at least one void spacing within the nucleation time; otherwise $\alpha_1 < 1$ throughout the accessible temperature range and no kinetic saturation occurs. For the parameter ranges in this work ($R_v \in \{1, 10, 25\} \text{ nm}$, $p \in [0.05, 0.20]$, $w_O \in [500, 3000] \text{ ppm}$), this condition is satisfied at all relevant temperatures.

Appendix C.6. Dependence of T_{peak} on the nucleation time t

Eq. (C.15) makes explicit that t enters the expression for T_{peak} only through a logarithm. Differentiating with respect to $\ln t$:

$$\frac{dT_{\text{peak}}}{d \ln t} = -\frac{E_m + E_f(v_{\text{N}}^{(\text{b})})}{k [\ln(A_0 t / \lambda_v^2)]^2} = -\frac{k T_{\text{peak}}^2}{E_m + E_f(v_{\text{N}}^{(\text{b})})}, \quad (\text{C.16})$$

where $A_0 \equiv f z \lambda^2 \nu / (6 c_{\text{O}})$. Thus increasing t lowers T_{peak} , but only weakly. For an order-of-magnitude change in t :

$$\Delta T_{\text{peak}} \approx -\frac{k T_{\text{peak}}^2}{E_m + E_f(v_{\text{N}}^{(\text{b})})} \ln 10. \quad (\text{C.17})$$

Taking $E_m + E_f(v_{\text{N}}^{(\text{b})}) = 5.19$ eV (intermediate stoichiometry) and $T_{\text{peak}} = 1448$ K (the expected value for $R_v = 1$ nm, $w_{\text{O}} = 1500$ ppm, $p = 0.05$) gives:

$$\Delta T_{\text{peak}} \approx -80 \text{ K}. \quad (\text{C.18})$$

This confirms that T_{peak} is only logarithmically sensitive to the nucleation time.

Appendix C.7. Numerical validation and parametric trends

Equation (C.14) was compared against direct numerical maximization of the full $|\Delta\sigma|(T)$ from Eq. (7) (including the $\text{O}_i^{(\text{s})}$ and $v_{\text{N}}^{(\text{s})}$ contributions) for a grid of parameter combinations at intermediate stoichiometry:

Table C.7: Comparison of T_{peak} from the closed-form expression (Eq. (C.14)) and from direct numerical maximization of the full $|\Delta\sigma|(T)$ (Eq. (7)), at intermediate stoichiometry.

	T_{peak} (formula) [K]	T_{peak} (numerical) [K]	Difference [K]
$R_v = 1$ nm, $w_{\text{O}} = 1500$ ppm, $p = 0.05$	1447	1448	< 1
$R_v = 1$ nm, $w_{\text{O}} = 1500$ ppm, $p = 0.20$	1416	1416	< 1
$R_v = 10$ nm, $w_{\text{O}} = 1500$ ppm, $p = 0.05$	1627	1627	< 1
$R_v = 25$ nm, $w_{\text{O}} = 1800$ ppm, $p = 0.05$	1721	1721	< 1
$R_v = 25$ nm, $w_{\text{O}} = 1800$ ppm, $p = 0.20$	1676	1677	< 1
$R_v = 25$ nm, $w_{\text{O}} = 1500$ ppm, $p = 0.05$	1712	1712	< 1

Agreement is within 1 K in all cases, confirming that the neglected $\text{O}_i^{(\text{s})}$ and $v_{\text{N}}^{(\text{s})}$ contributions shift the peak negligibly for the DFT parameter set used here.

Equation (C.15) yields the following parametric trends, consistent with Fig. 6:

- *Increasing R_v* increases λ_v , decreasing the logarithm and therefore increasing T_{peak} . The ~ 180 K upward shift from $R_v = 1$ nm to $R_v = 10$ nm reported in Section 3.3 is reproduced quantitatively.
- *Increasing p* decreases λ_v and slightly decreases T_{peak} . The effect is weak because p enters through a logarithm.
- *Increasing c_{O}* increases the denominator of the logarithm's argument and slightly increases T_{peak} ; again, the effect is logarithmic and weak.
- *Stoichiometry* enters the numerator through $E_f(v_{\text{N}}^{(\text{b})})$: N-rich conditions (higher E_f) shift T_{peak} upward.

Appendix C.8. Consequence for the porosity dependence of $|\Delta\sigma|$

The prefactor of Eq. (7) contains the factor $(1-p)/p$, which varies by nearly $4.75\times$ between $p = 5\%$ and $p = 20\%$. Yet the peak $|\Delta\sigma|$ varies far more weakly (Fig. 8). This subsection explains why.

From Eq. (37), the peak $|\Delta\sigma|$ (evaluated at T_{peak} where $\alpha_1 = 1$ by Eq. (C.13)) can be written as:

$$|\Delta\sigma|_{\text{peak}} = \frac{8}{3} \frac{c_{\text{O}}}{a^3} \underbrace{\frac{(1-p)}{p} R_v}_{\eta'} \times \underbrace{\frac{[\text{N}_{\text{N}}^{(\text{s})}]}{[\text{N}_{\text{N}}^{(\text{b})}]}}_{\text{from } g(T)} \times \exp\left(\frac{E_{\text{seg}}}{kT_{\text{peak}}}\right) \times |E_{\text{ad}}(T_{\text{peak}})|, \quad (\text{C.19})$$

where $[\text{N}_{\text{N}}^{(\text{s})}] = \frac{3}{2}(p/(1-p))(a/R_v)$ from Eq. (28). The product of the two underbraced factors simplifies as:

$$\frac{(1-p)}{p} R_v \times \frac{\frac{3}{2} \frac{p}{1-p} \frac{a}{R_v}}{[\text{N}_{\text{N}}^{(\text{b})}]} = \frac{3a/2}{[\text{N}_{\text{N}}^{(\text{b})}]}, \quad (\text{C.20})$$

where the factors $(1-p)/p$ and $p/(1-p)$ cancel exactly, and R_v cancels as well. The only remnant is $1/[\text{N}_{\text{N}}^{(\text{b})}] = 1/(1 - [\text{N}_{\text{N}}^{(\text{s})}])$, which is close to unity because $[\text{N}_{\text{N}}^{(\text{s})}] \ll 1$ for the void sizes of interest. Substituting back:

$$|\Delta\sigma|_{\text{peak}} = \frac{4c_{\text{O}}}{a^2} \times \frac{1}{[\text{N}_{\text{N}}^{(\text{b})}]} \times \exp\left(\frac{E_{\text{seg}}}{kT_{\text{peak}}}\right) \times |E_{\text{ad}}(T_{\text{peak}})|. \quad (\text{C.21})$$

The prefactor $4c_{\text{O}}/a^2$ is independent of both p and R_v . Thus, the peak $|\Delta\sigma|$ depends on porosity only through two weak channels:

- (a) The thermodynamic residual $1/[\text{N}_{\text{N}}^{(\text{b})}]$ deviates from unity only when $[\text{N}_{\text{N}}^{(\text{s})}] = \frac{3}{2}(p/(1-p))(a/R_v)$ is non-negligible. For $R_v = 25$ nm, this residual varies by less than 1% across $p = 5\text{--}20\%$; for $R_v = 1$ nm, it varies by 18%.
- (b) Increasing p reduces λ_v (voids are closer together), so the kinetic saturation condition $\sqrt{D_1 t} = \lambda_v$ is met at a lower temperature. From Eq. (C.15):

$$T_{\text{peak}} \sim \frac{1}{\ln(\dots p^{2/3} \dots)}, \quad (\text{C.22})$$

so T_{peak} responds only logarithmically to changes in p . Over $p = 5\text{--}20\%$, the downward shift is only 31 K for $R_v = 1$ nm and 44 K for $R_v = 25$ nm. This lower T_{peak} increases $\exp(E_{\text{seg}}/kT_{\text{peak}})$ (stronger segregation drive at lower temperature), which slightly raises $|\Delta\sigma|_{\text{peak}}$.

Table C.8 quantifies both effects for $w_{\text{O}} = 1500$ ppm at intermediate stoichiometry. The variation column reports the total change in $|\Delta\sigma|_{\text{peak}}$ (from Eq. (C.21)) relative to $p = 5\%$. For $R_v = 25$ nm (the conditions of Fig. 8), the total change in peak $|\Delta\sigma|$ from $p = 5\%$ to $p = 20\%$ is only 8%, almost entirely from channel (b). For $R_v = 1$ nm, the total change is 26%, with 18% from channel (a) and 7% from channel (b). In both cases, the peak $|\Delta\sigma|$ changes by at most a factor of 1.3 over a fourfold increase in porosity, far smaller than the factor of 4.75 that the naïve $(1-p)/p$ scaling would suggest.

Table C.8: Porosity dependence of the peak $|\Delta\sigma|$ at two void radii ($w_{\text{O}} = 1500$ ppm, intermediate stoichiometry). Each row is evaluated at that porosity's own T_{peak} from Eq. (C.14).

	p [%]	T_{peak} [K]	$1/[N_{\text{N}}^{(\text{b})}]$	$ \Delta\sigma $ variation
$R_v = 1$ nm	5	1448	1.040	—
	10	1432	1.088	+8%
	15	1423	1.148	+17%
	20	1416	1.223	+26%
$R_v = 25$ nm	5	1713	1.002	—
	10	1690	1.003	+4%
	15	1678	1.005	+7%
	20	1669	1.007	+8%

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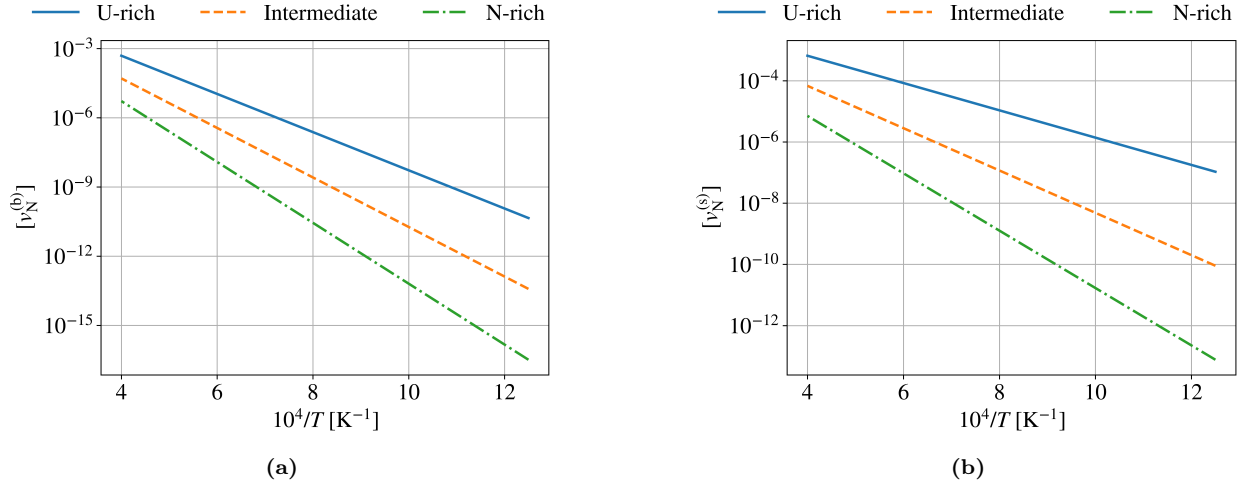


Figure D.9: (a) Concentration of N vacancies in bulk UN calculated from the formation energies in Table 1. (b) Concentration of N vacancies on the UN (001) surface calculated from the formation energies in Table 1.

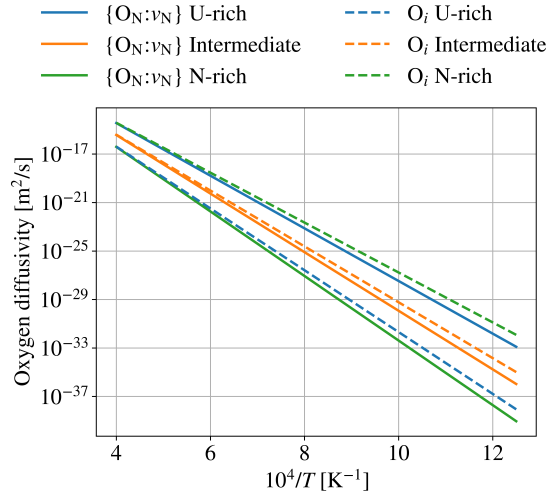


Figure D.10: (Color online) Diffusivity of oxygen impurities in bulk UN by the defects $\{O_N:v_N\}$ and O_i under U-rich (blue), Intermediate (orange), and N-rich (green) conditions.

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